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Truthful online bundle double auction mechanisms for a transport market with dynamic demands and supplies

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ABSTRACT

In this paper, we study the transaction mechanism design problem for a market where shippers and carriers dynamically arrive at and depart from an online freight platform to trade their transport services over a transport network. The agents are self-interested and their valuations, costs, and arrival and departure times are private information. We propose online double auction mechanisms to dynamically match the demands and supplies and set the transaction prices for three different cases. We first consider the case that each shipper wants to procure one unit of transport service on a lane and each carrier can supply multiple units of transport service on multiple lanes, and propose the Online Bundle Double Auction (OBDA) mechanism for it. We show that the mechanism is *feasible, incentive-compatible*, *individually rational*, and *budget-balanced*. Then we further consider two more general cases: the case where the shippers have multi-unit demand on one lane and the case where the shippers have multi-unit indivisible demand on multiple lanes. We show that the framework of the OBDA mechanism can still be applied to these cases by modifying some rules. We show that the two modified mechanisms still have good properties. Moreover, we conduct numerical studies to investigate the impacts of some parameters on these mechanisms' performance.

1. Introduction

Road freight transport is important for national, regional, and municipal economies (Lafkihi et al., 2019; Park et al., 2023; Peng et al., 2024). In China, the volume of road freight transport reaches 37.12 billion tons in 2022, accounting for about 73.34 % of the gross freight volume (National Bureau of Statistics, 2023). In Europe, road transport accounts for about a quarter of freight transport performance in 2022 (Eurostat, 2023a). However, it also has suffered from inefficiency because of the imbalance geographic distribution of freight and information asymmetry between supply and demand (Guo et al., 2022). From Eurostat (2023b), the ratio of empty vehicles ranges from 15 % to 30 % in most European member States and reaches 40.4 % in Cyprus. To alleviate these adverse effects, many online freight platforms (e.g., www.for-u.com; www.fulltruckalliance.com, also called third-party exchange service providers, car-free carrier platforms) have developed to collect cargo and vehicle information and then match them (Liang et al., 2020; Lau and Li, 2021; Guo et al., 2022; Park et al., 2023). According to statistics, these platforms have significantly reduced transport costs and carbon emissions by reducing empty travel miles and raising the vehicle utilization rate (Guo et al., 2022).

The operation of the online freight platforms involves dynamically collecting transport demand and supply information, matching

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the demand (cargoes) with the supply (vehicles), and setting transaction prices (Mancini et al., 2021; Tian et al., 2022). Most of the existing literature assumes they act as centralized decision-makers to match the demand and supply directly according to information submitted by shippers and carriers (Fernández and Sgalambro, 2020). However, this ideal centralized scheme is generally unrealistic in bilateral transport markets since agents (shippers and carriers) are generally autonomous players, even competitors and they are not willing to share their private information with others (Gansterer et al., 2020). Given that the transport costs and valuations are private information (Wei, 2023), some experts try to design mechanisms to yield individual and collective desirable schemes with incomplete information sharing from self-interested players (e.g., Huang and Xu, 2013; Lyu et al., 2019; Guo et al., 2022). Major of the works are based on one-sided auctions which need to be implemented many times (e.g., Figliozzi, 2006; Lyu et al., 2019), and may result in very high transaction costs (Chu, 2009; Guo et al., 2022). In addition, they overlook the profit of third-party auctioneers (platforms). To overcome these shortages, some works (e.g., Huang and Xu, 2013; and Yu et al., 2022) propose double auction-based transport service transaction mechanisms where shippers and carriers simultaneously bid first, and then the platform allocates the transactions and determines the transaction prices. Sun et al. (2019) and Guo et al. (2022) further consider the impacts of transaction costs on the design of double auction mechanism for transport service transactions. Double auction mechanisms are also investigated for the secondary transport market where some carriers want to subcontract their unprofitable transport requests to others (e.g., Handoko and Lau, 2016; Xu et al., 2016).

Up to now, most of the transaction mechanisms for transport services consider only the static situation where all the shippers and carriers arrive and trade their transport services in one period. However, the transport service transaction is highly dynamic and random in practice (Xu and Huang, 2013; Regnier-Coudert et al., 2016; and Scherr et al., 2023). The agents usually arrive at the platform dynamically, stay for some periods, and then leave the market even if they do not get the transactions. Shippers' demands and carriers' supplies are only effective in these specific periods (Li et al., 2016; Mancini et al., 2021; Park et al., 2023). For example, on some online freight platforms (e.g., FOR-U Truck, Full Truck Alliance), shippers will give loading time information, i.e., the latest departure time, when posting transport requests. These requests are only valid until the departure time. Similarly, the carriers are time-sensitive with respect to transport services as their transport schedules are subject to the timing of transport requests (Xu and Huang, 2014; Deng et al., 2021). For example, the shipments of some transport companies and truck fleets are scheduled in advance and cannot changed (e.g., freight shuttle, cargo liner, etc.). Individual drivers usually have limited patience to wait for the results of the transaction. Once they lose their patience, they would rather return empty (Farren et al., 2024). The agents' arrival time and departure time are usually their private information and they have no incentive to share them truthfully. Some agents may delay arrival and observe market trends to strategically wait for a more preferable price. In addition, they may post an earlier departure time to get a deal as soon as possible. These strategic behaviors would make the actual transaction outcome deviate from the optimal one, thus reducing market efficiency. Therefore, how to match the demands and supplies efficiently considering their dynamic arrivals and departures under incomplete information is a very important problem faced by the online freight platforms.

A straightforward idea to solve the problem is to repeat static double auction mechanisms to periodically clear the market (Xu and Huang, 2013). However, this idea ignores the agents' private information about their arrival and departure times. As pointed out by Yu et al. (2024), the agents may strategically lose money in the current period to pursue more favorable prices in the next periods. Thus the mechanisms are hardly incentive-incompatible. Considering such private information, Kong et al. (2021) designed an online double auction mechanism where the agents dynamically arrive and leave to trade transport services on one lane. Generally, shippers' demands are heterogeneous, and they require transport services on different lanes. Also, carriers prefer to offer a bundle of transport services on different lanes to capture the benefits of the scope economy (Song and Regan, 2005; Xu and Huang, 2014; Li et al., 2023). To our best knowledge, there is no mechanism considering simultaneously the agents' dynamic arrives and departures, private information on arrivals and departures, and bundled transport service transactions. To fill the gap between theory and practice, this study tries to design some online bundle double auction mechanisms for dynamic transport service transactions considering these factors.

In this paper we consider a transport market where some carriers provide Full Truck Loading (FTL) transport services over a transport network. The shippers and carriers will arrive at and depart from the online freight platform dynamically to trade their transport services over the network. The carriers will supply a bundle of transport services.[1] The shippers' demands have three cases: (i) the basic one that the shippers procure one unit of transport service on a lane; (ii) an extension that the shippers procure multiple divisible units on a lane; (iii) another extension that the shippers procure multi-unit indivisible demand on multiple lanes from one carrier. All the agents are self-interested and their valuations, costs, arrival, and departure times are their private information. Here we design the mechanisms with the properties of *feasibility* (exactly allocate the transactions among the agents), *individual rationality*, *incentive compatibility*, *budget balance*, and *high mechanism efficiency* for these three cases.

As we know, it is not easy to design transaction mechanisms with the expected properties for our problems. The first difficulty is to ensure incentive compatibility. Note that not only the agents' valuations and costs but also their arrival and departure times are private information in dynamic transactions, so it is not easy to ensure that the agents reveal their information truthfully. The second one is to ensure the *weak* budget balance, that is, at no time should the platform pay more than it has received. This is because the agents may trade multiple units of transport services on different lanes, or even a bundle of transport services at once. The third one is to ensure

[1] Generally, the FTL demand and supply types can be characterized by the trading quantity (unit), item (transport service on different lanes), whether to be divisible (i.e., partially satisfied), and whether to be satisfied by one agent (Huang and Xu, 2013; Xu et al., 2016; Guo et al., 2022). Most of the state-of-the-art online double auction mechanisms considers that the demands and the supplies are one/multiple units of one item (e.g., Li et al., 2019; Dibaj et al., 2020; Kong et al., 2021; Patel et al., 2021). We give a more detailed literature review in Section 2.

that the online mechanism has high efficiency with dynamic and stochastic demands and supplies. In the static trading environment, the transactions are conducted at once and the current transaction results do not affect the decisions afterward. In this study, the auctioneer does not know the arrival of the agents in the future when it makes the current transaction decisions (Li et al., 2019).

To address these problems, we first propose the **O**nline **B**undle **D**ouble **A**uction (OBDA) mechanism for the basic case (i). This mechanism consists of the pre-determined pricing rule, admission rule, allocation and payment rules, and survival rule. Particularly, we construct a pre-determined pricing rule based on the static Bundle Double Auction (BDA) mechanism (Xu et al., 2016). To effectively incentivize the agents to reveal their arrival and departure times truthfully, we design an admission rule for newly arrived agents that takes into account the pricing schemes for the periods before the arrivals of the agents and determines the monotonically increasing (decreasing) admission prices for shippers and carriers over periods. A similar monotonous payment rule is designed to prompt the mechanism to balance its budget and to incentivize the agents to reveal their valuation and cost information truthfully. In addition, we design a survival rule to improve efficiency by retaining some unfilled agents due to supply–demand imbalance. We can show that the OBDA mechanism is *feasible*, *incentive-compatible*, *individually rational*, and *budget-balanced*.

Then, we consider the extension case (ii) and case (iii). For the former case, it is difficult to ensure incentive compatibility for the agents with multi-unit divisible demand or supply even in the static bundled transport service transactions (Xu et al., 2016; Sun et al., 2019). To overcome this difficulty, we use the idea of the "padding" method (Chu, 2009; Guo et al., 2024b) to modify the OBDA mechanism and propose the OBDA-padding mechanism. For the latter case, we use the idea of the "*single matching pattern*" to modify the pre-determined rule of the OBDA mechanism and propose the OBDA-Single Matching Pattern (OBDA-SMP) mechanism. We show that the two modified mechanisms still have good properties (*feasibility*, *individual rationality*, *individual rationality*, and *budget balance*). The OBDA-SMP mechanism can be applied to the other two cases but has less efficiency than the other two mechanisms. Moreover, we conduct numerical studies to compare the efficiencies of the three mechanisms and explore the impacts of the market size and the agents' patience on their performance.

We organize the rest of the paper as follows: In Section 2 we review relevant literature to identify the research gap and position our study. In Section 3 we introduce the problem and discuss the assumptions. We design the OBDA mechanism for the basic case in Section 4. In Section 5 we propose the OBDA-padding and OBDA-SMP mechanisms for two general multi-unit demand cases. In Section 6 we present the numerical studies to examine the impacts of some parameters on the mechanisms' performance. Finally, in Section 7, we conclude the paper and suggest topics for future study. We present all of the proofs in the Appendix.

2. Literature review

In this paper, we study the double auction-based transport service transaction mechanism design problem considering dynamic demands and supplies. There are three streams of literature related to this work, namely the dynamic vehicle-cargo matching problem, the auction-based mechanisms for transport service transactions, and the online double auctions.

The first stream of literature concerns the dynamic vehicle-cargo matching problems where a carrier or third-party agency efficiently matches dynamically arriving demands (cargoes) and supplies (vehicles) under complete information. Most of the papers investigate the vehicle-cargo matching problem with time window constraints (e.g., Fernández and Sgalambro, 2020; Lau and Li, 2021; and Mancini et al., 2021). Regnier-Coudert et al. (2016) developed a constructive solver to assign trucks and trailers to a set of dynamically arriving and updating transport requests in real time. Deng et al. (2021) used the dynamic Bayesian network model to predict the multi-time sequence matching probability of vehicle-cargo matching. Considering the degrees of attribute matching and environmental influence, Tian et al. (2022) designed a vehicle-cargo matching algorithm based on the improved dynamic Bayesian network for the case where the availability of vehicles and cargoes vary at different periods. Our study also involves the efficient dynamic vehicle-cargo matching problem. Different from the above works, we consider the case that the agents have private information on their valuations, costs, arrival time, and departure time and we design auction-based mechanisms to make the equilibrium matching outcome close to the optimal one.

The second stream of literature concerns the auction-based mechanisms for transport service transactions. Lafkihi et al. (2019) systematically reviewed the literature of freight transport service procurement mechanisms, most of which are auction-based mechanisms. There are two themes in this stream, one-sided auction-based mechanisms and double auction-based mechanisms. The former enables many-to-many trading in transport markets by implementing one-sided auctions (forward or reverse auctions) multiple times. Li et al. (2015) proposed an exchange approach for carrier collaboration and suggested the Vickrey-Clarke-Groves (VCG) auction to allocate profits among the carriers. Based on the clock-proxy auction, Chen (2016) designed an exchange mechanism for the carrier collaboration problem in less-than-truck load transport. Lai et al. (2017) proposed an iterative auction mechanism for the FTL transport service transactions among carriers, which repeatedly uses the second-price auction to exchange transport requests. Lyu et al. (2019) proposed an iterative exchange mechanism for the case where the carriers trade multiple bundles of transport services.

To design more time-efficient and practically attractive mechanisms than one-sided auction-based mechanisms, some studies propose transaction mechanisms based on double auctions. Huang and Xu (2013) proposed some double auction mechanisms for multi-unit transport service transactions in logistics e-marketplaces. Xu et al. (2016) designed three bundle double auction mechanisms based on the trade reduction approach for transport service transactions among carriers over a transport network. Sun et al. (2019) proposed bundle double auction mechanisms for intermodal transport service transactions. Liang et al. (2020) designed double auction mechanisms where the carriers can offer quantity discounts. Guo et al. (2022) designed multi-unit double auction mechanisms considering transaction costs and private demand information. Yu et al., (2022) proposed multi-unit double auction mechanisms for attributes transport service transactions.

The above works focus on the static situation, where all the shippers and carriers trade their transport services in one period. Considering random arrival transport demands, Figliozzi (2006) developed a sequential second-price auction mechanism to trade transport services. Scherr et al. (2023) proposed an auction-based collaboration mechanism with dynamic request acceptance. Guo et al. (2024a) designed an online auction-based transaction mechanism for instant delivery platforms to handle dynamic requests with multi-dimensional requirements. The above works, focus on one-sided auctions. Xu and Huang (2013) devised a continuous double auction mechanism for transport service transactions with stochastic demand and supply. However they did not consider the times that the agents stay in the market and assumed that the arrival and departure information is common knowledge. Kong et al. (2021) proposed an online double auction mechanism for the transactions of the on-demand logistics service on one lane. Different from them, we consider the case where the agents can trade bundles of transport services on different lanes.

The third stream of literature concerns the applications of online double auctions in other areas, e.g., electricity trading, on-demand multi-party carpooling, and crowdsourcing trading. The online double auctions model the dynamic bilateral trading problem with private information where the self-interested buyers (shippers) and sellers (carriers) arrive and depart dynamically. Most of the literature focuses on the case where the agents trade a homogenous item. Bredin et al. (2007) first provided a framework that allows a truthful online double auction mechanism to be constructed from a truthful static double auction mechanism. Wang et al. (2010) designed a truthful online double auction mechanism for spectrum allocation, where the secondary users arrive following a Poisson distribution and want to lease for different time-slots of spectrum occupancy from the primary users. Miyashita (2014) proposed an online multi-unit double auction for perishable goods to reduce trade failures. An et al. (2017) proposed a truthful online double auction mechanism for the multi-unit electricity trading of energy within the smart grid. To protect users' sensitive information, Li et al. (2019) designed privacy-preserving online double auction mechanisms. Dibaj et al. (2020) proposed a cloud priority-based online double auction mechanism, considering the perishability and time constraints in Infrastructure-as-a-Service (IaaS) secondary markets. Yu et al. (2024) designed two double auction mechanisms for on-demand multi-party carpooling which integrate frustration-based promotion.

Some papers consider the case where the agents trade heterogeneous items. Patel et al. (2021) proposed an online double auction mechanism for IaaS clouds where the agents have different resource requirements or availability. Ding et al. (2023) designed online double auction mechanisms for on-demand integrated ride-sourcing platforms on which travelers have travel requirements on multiple lanes and transport service providers own multiple vehicles. However, these two works did not consider the agents' misrepresentation of their arrival and departure times. Assuming that these times are the agents' private information, Liu et al. (2022) designed online double auctions for mobile crowdsourcing where the agents can submit multiple bundle bids containing multiple types of tasks. They assume that an agent has demand or supply for each period when it stays in the market and their mechanisms are not truthful in our situation.

Different from this stream of the literature, our work has the following distinctions: (i) the agents can trade heterogeneous items (transport services on multiple lanes), (ii) the agents can bid for a bundle of items (transport services over the transport network), and (iii) agents have limited demand and supply for their staying periods in the market.

3. Problem setting and assumptions

3.1. Problem description

Consider a FTL transport market with multiple shippers, multiple carriers, and an online freight platform. Carriers provide transport services using homogeneous vehicles over a transport network with multiple origin–destination pairs (called lanes). Let L denote the set of lanes. Shippers want to procure transport services on some lanes from these carriers on the online freight platform.[2] The platform provides transaction services between shippers and carriers. In general, we refer to the shippers and carriers as "agents".

Consider discrete time periods $T = \{0, 1, 2, ...\}$ over which the shippers dynamically arrive at the platform to trade the transport services. Let a_i be shipper i's arrival time, which is referred to as the time when shipper i is aware of its desire to trade on the platform. Shipper i stays at the platform for some periods and then leaves. Let d_i be shipper i's departure time, which is referred to as the latest period when it can accept the transaction. During its stay periods $[a_i, d_i]$, shipper i wants to procure q_i units of transport services on a targeted lane $l_i \in L$ with marginal valuation v_i.[3] We first consider the case that a shipper's demand is "divisible", i.e., the demand can be partially met and can be met by more than one carrier. In Subsection 5.2 we shall consider the case that the shipper procures multiple units of transport services on multiple lanes from only one carrier.

The carriers also arrive at and depart from the platform dynamically over the T periods. Let a_j and d_j be carrier j's arrival and departure times, respectively. During its stay periods $[a_i, d_i]$, carrier j wants to sell a bundle of transport services $b_j \in B$ with cost c_j, where B is the set of all the possible supply bundles. Each bundle $b \in B$ consists of $q^l(b)$ units of transport services on lane $l \in L$. We assume that a carrier's bundle cannot be partially sold, but can be allocated to different shippers. This is because carriers usually

[2] In our study, the scope of shippers is not limited to original shippers in the primary transport market, but can also be carriers who want to outsource their subcontract transport requests in the secondary transport market, as discussed by Handoko and Lau (2016) and Xu et al. (2016).

[3] Our basic case, discussed in Section 4, which the shippers procure one unit of transport service on a lane, is a special case of the case with multi-unit demand on one lane. For simplicity of presentation, we directly describe the multi-unit demand case in Section 3. We further describe in Subsection 5.2 the different settings in the case with multi-unit indivisible demand on multiple lanes differently from those in the multi-unit demand case.

provide transport services on several fixed and familiar lanes (e.g. pairs of open lanes, multi-point circulation lanes), and selling its bundle partially may result in increased empty travels.

Then shipper i has the type $\theta_i = \{a_i, d_i, l_i, q_i, v_i\} \in \Theta_I$ and carrier j has the type $\theta_j = \{a_j, d_j, b_j, c_j\} \in \Theta_J$, where Θ_I is the set of shippers' possible types and Θ_J is the set of carriers' possible types. Fig. 1 shows a simple example where there are six periods of transactions. In this example, shipper i_1 arrives at the market in period 0 and will leave after period 2, and it wants to buy two units of transport service on lane m with a marginal valuation of 2 during this time. Note that there is a time window requirement for each transport, we assume that all of the agents are boundedly patient and stay in the market for at most $K+1$ periods, i.e., $d_i \le a_i + K$ for any shipper i and $d_j \le a_j + K$ for any carrier j. Generally, the agents' arrival times, departure times, and valuations (costs) are their private information. We assume that the shippers' demand (i.e., lane l_i and quantity q_i) and the carriers' supply (i.e., bundle b_j) are common knowledge. This assumption is common in many studies on transport service transactions (Huang and Xu, 2013; Xu et al., 2016; Sun et al., 2019, Yu et al., 2022).

The online freight platform, as the third-party auctioneer, holds the online bundle double auction to allocate the transactions and determine the transaction prices for both shippers and carriers in each period. In each period, the newly arrived shipper i bids its type $\widehat{\theta}_i = \{\widehat{a}_i, \widehat{d}_i, l_i, q_i, \widehat{v}_i\}$ and the newly arrived carrier j asks its type $\widehat{\theta}_j = \{\widehat{a}_j, \widehat{d}_j, b_j, \widehat{c}_j\}$ to the platform, where $\widehat{a}_i$, $\widehat{d}_i$, and $\widehat{v}_i$ are the arrival time, departure time, and bid price submitted by shipper i, respectively. The notations for a carrier are similar with j instead of i and $\widehat{c}_j$ is carrier j's ask price, respectively. Note that in practice, the arrival time can be recorded automatically according to the agent's bidding time, which is not necessarily submitted. Each agent is self-interested and submits its bid or ask to maximize its own utility. So it may misrepresent its type, i.e., $\widehat{\theta}_i \neq \theta_i$ or $\widehat{\theta}_j \neq \theta_j$, when it finds that lying is beneficial. We suppose that a shipper does not arrive early ($\widehat{a}_i \ge a_i$) or leave late ($\widehat{d}_i \le d_i$). The former is consistent with our interpretation of the arrival time. As for the latter, trading after d_i is unacceptable for shipper i because its utility will be zero. We make similar assumptions for the carriers. In addition, we assume that there is no collusion among the agents.

Remark 1 We assume that the agents have an incentive to arrive late and leave early. Although some transport companies and truck fleets, as we mentioned in Section 1, will voluntarily post the real latest departure information because the shuttles are usually fixed for long periods. However, individual truckers, with private information on their latest departure, still dominate the transport market. For example, they account for about 78.22 % in China (CFLP, 2024). We therefore consider this more general assumption that all carriers may misrepresent their departure time.

Suppose that shipper i's utility is the difference between the valuation and payment if it trades in period $t \in [a_i, d_i]$ and carrier j's utility is the difference between the payoff and cost if it trades in period $t \in [a_j, d_j]$. Otherwise its utility is zero. The platform's profit is the difference between the total payment from shippers and the total payoff to carriers. The social welfare is the sum of the agents' utilities and the platform's profit.

3.2. Desiderata

Note that the revelation principle still holds for online mechanisms (Bredin et al., 2007). So we consider the direct-revelation online bundle double auction mechanism where each agent sends to the auctioneer a single direct claim about its type (Myerson, 1981). We also consider "closed" auctions where an agent receives no feedback before submitting its type so that it cannot condition its strategy on other agents' bids and asks.

The online bundle double auction mechanism consists of an allocation rule that specifies the number of each agent's trade units, and a payment rule that specifies the money for each trading agent to pay or receive. Let $M = \{(x, y), (\pi, \omega)\}$ be an online double auction mechanism that uses the allocation rule $(x, y) = \left\{ \left(x_i^t\right)_{i \in I^{\le t}}, \left(y_j^t\right)_{j \in J^{\le t}} \right\}^{t \in T}$ and the payment rule $(\pi, \omega) = \left\{ \left(\pi_i^t\right)_{i \in I^{\le t}}, \left(\omega_j^t\right)_{j \in J^{\le t}} \right\}^{t \in T}$ where $I^{\le t}(J^{\le t})$ is the set of the shippers (carriers) with arrivals no later than period t. Let $\widehat{\theta}_{I^{\le t}} = \{\widehat{\theta}_i | i \in I^{\le t}\}$ be the set of the types submitted by all shippers with arrivals no later than period t. The notation for the carriers is similar with J instead of I. Given $\widehat{\theta}_{I^{\le t}}$ and $\widehat{\theta}_{J^{\le t}}$, $x_i^t(\widehat{\theta}_{I^{\le t}}, \widehat{\theta}_{J^{\le t}}) \in \{0, 1, \ldots, q_i\}$ indicates the number of trading units of shipper i in period t and $y_j^t(\widehat{\theta}_{I^{\le t}}, \widehat{\theta}_{J^{\le t}}) \in \{0, 1\}$ indicates whether or not carrier j sells its bundle in period t. $\pi_i^t(\widehat{\theta}_{I^{\le t}}, \widehat{\theta}_{J^{\le t}}) \in \mathbb{R}_0^+$ indicates the payment by shipper i and $\omega_j^t(\widehat{\theta}_{I^{\le t}}, \widehat{\theta}_{J^{\le t}}) \in \mathbb{R}_0^+$ indicates the revenue received by carrier j. Then a feasible online bundle double auction mechanism must satisfy the following conditions: (i)

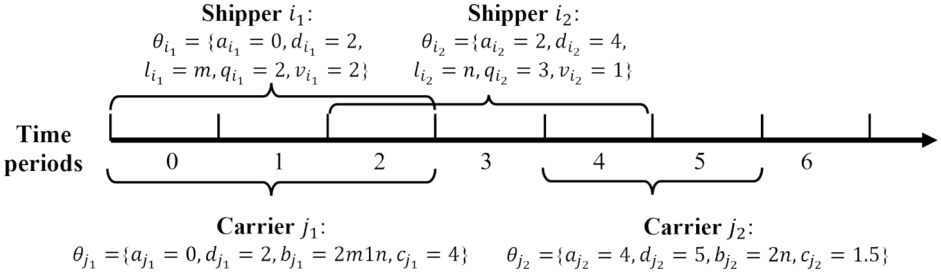


Fig. 1. A simple example.

$\sum_{t\in[\widehat{a}_i,\widehat{d}_i]} x_i^t \le q_i$; (ii) $\sum_{t\in[\widehat{a}_j,\widehat{d}_j]} y_j^t \le 1$; (iii) $x_i^t = 0$ for any shipper i and $t \notin [\widehat{a}_i, \widehat{d}_i]$; and (iv) $y_j^t = 0$ for any carrier j and $t \notin [\widehat{a}_j, \widehat{d}_j]$.

The notations used in this paper are listed in Table 1.

A good online mechanism should possess the following properties: *incentive compatibility*, *strong feasibility*, *ex-post weak budget balance*, and *ex-post individual rationality*.

Incentive compatibility requires that no misreporting can increase an agent's utility, regardless of what other agents bid and what the future dynamics are:

Definition 1 (Incentive compatibility). *An online bundle double mechanism $M = \{(x, y), (\pi, \omega)\}$ is incentive-compatible or truthful if the followings hold for any $\widehat{\theta}_i \in \Theta_I$ and $\widehat{\theta}_j \in \Theta_J$:*

(I) for any shipper i,

$$\sum_{t\in[a_i,d_i]} x_i^t(\{\theta_i\} \cup \widehat{\theta}_{I^{\le t}\setminus\{i\}}, \widehat{\theta}_{J^{\le t}})v_i - \sum_{t\in[a_i,d_i]} \pi_i^t(\{\theta_i\} \cup \widehat{\theta}_{I^{\le t}\setminus\{i\}}, \widehat{\theta}_{J^{\le t}}) \ge \sum_{t\in[\widehat{a}_i,\widehat{d}_i]} x_i^t(\widehat{\theta}_{I^{\le t}}, \widehat{\theta}_{J^{\le t}})v_i - \sum_{t\in[\widehat{a}_i,\widehat{d}_i]} \pi_i^t(\widehat{\theta}_{I^{\le t}}, \widehat{\theta}_{J^{\le t}})$$

(II) for any carrier j,

$$\sum_{t\in[a_j,d_j]} \omega_j^t\big(\widehat{\theta}_{I^{\le t}}, \{\theta_j\} \cup \widehat{\theta}_{J^{\le t}\setminus\{j\}}\big) - \sum_{t\in[a_j,d_j]} y_i^t\big(\widehat{\theta}_{I^{\le t}}, \{\theta_j\} \cup \widehat{\theta}_{J^{\le t}\setminus\{j\}}\big)c_j \ge \sum_{t\in[\widehat{a}_j,\widehat{d}_j]} \omega_j^t(\widehat{\theta}_{I^{\le t}}, \widehat{\theta}_{J^{\le t}}) - \sum_{t\in[\widehat{a}_j,\widehat{d}_j]} y_j^t(\widehat{\theta}_{I^{\le t}}, \widehat{\theta}_{J^{\le t}})c_j$$

Table 1
The notations used in this paper.

Notations	Meanings
• Sets	
L	The set of lanes
Ω	The discrete set of the platform's operating periods
$I^{\le\tau}(J^{\le\tau})$	The set of the shippers (carriers) with arrivals no later than period τ
$\widehat{\theta}_{I^{\le\tau}}(\widehat{\theta}_{J^{\le\tau}})$	The type set of all the shippers (carriers) with arrivals no later than period τ
$\overline{I}^{\tau}(\overline{J}^{\tau})$	The set of active shippers (carriers) in period τ
$I(J)$	A set of shippers (carriers)
$B(O)$	The set of all the possible transport supply (demand) bundles
E	The set of all the possible matching pattern
• Parameters	
i	Shipper index
j	Carrier index
l	Lane index,$l \in L$
τ	Period index,$\tau \in \Omega$
$a_i(a_j)$	The arrival time of shipper i (carrier j)
$d_i(d_j)$	The departure time of shipper i (carrier j)
$\widehat{a}_i(\widehat{a}_j)$	The arrival time submitted by shipper i (carrier j)
$\widehat{d}_i(\widehat{d}_j)$	The departure time submitted by shipper i (carrier j)
K	The agents' patience boundary
$v_i(\widehat{v}_i)$	Shipper i's valuation (bid price) for each unit of transport service
q_i	Shipper i's demand of FTL transport services
l_i	The lane on which shipper i wants to procure transport services
$c_j(\widehat{c}_j)$	Carrier j's cost (ask price) for its transport service bundle
b_j	The transport service bundle that carrier j can provide
$q^l(b)$	The number of transport services on lane l included in transport service bundle b
$\theta_i(\theta_j)$	The type of shipper i (carrier j)
$f_I^{b_i}(I,J)$	Given I and J, the pre-buying price per unit of transport service for shipper i
$f_J^{b_j}(I,J)$	Given I and J, the pre-selling price for carrier j's supply bundle
$p_i^+(p_j^-)$	The threshold price for shipper i (carrier j)
$\pi_i^\tau(\omega_j^\tau)$	Shipper i's payment (carrier j's revenue) in period τ
S_l	The padding for lane l
$\boldsymbol{S}$	The supply padding vector
bt_i	The bidding time of shipper i
o_i	The transport service bundle that shipper i wants to procure
e	Matching pattern
$q^o(b,e)$	The number of demand bundle o matched by supply bundle b_j under matching pattern e
• Decision variables	
α_i	The number of trading units of shipper i
β_j	Whether carrier j trades
x_i^τ	The number of trading units of shipper i in period τ
y_j^τ	Whether carrier j sells its transport service bundle in period τ
z_{ij}^τ	Whether shipper i procures transport services from carrier j in period τ
$\alpha_i(e)(\beta_j(e))$	Whether shipper i (carrier j) trades under matching pattern e
$\gamma(e)$	Whether matching pattern e is selected

Strong feasibility requires that the total demand and supply on each lane should be balanced in any period:

Definition 2 (Strong feasibility). *An online bundle double mechanism* $M = \{(x,y),(\pi,\omega)\}$ *is strongly feasible if* $\sum_{i\in\{i|l_i=l,i\in I^{\le t}\}} x_i^t = \sum_{j\in J^{\le t}} y_j^t q^l(b_j)$, $\forall l \in L, \forall t \in T$.

The budget balance requires that the platform has a cash surplus in each period:

Definition 3 (*Ex-post weak* budget balance). *An online bundle double mechanism* $M = \{(x,y),(\pi,\omega)\}$ *is ex post weakly budget-balanced if* $\sum_{i\in I^{\le t}} \pi_i^t - \sum_{j\in J^{\le t}} \omega_j^t \ge 0$, $\forall t \in T$.

Individual rationality requires that an agent's utility is non-negative when it tells the truth regardless of other agents' reports:

Definition 4 (*Ex-post* individual rationality). *An online bundle double mechanism* $M = \{(x,y),(\pi,\omega)\}$ *is ex post individually rational if the following hold for any* $\theta_i \in \Theta_I$ *and* $\theta_j \in \Theta_J$*:*

(I) for any shipper i, $\sum_{t\in[a_i,d_i]} x_i^t(\{\theta_i\} \cup \widehat{\theta}_{I^{\le t}\setminus\{i\}}, \widehat{\theta}_{J^{\le t}})v_i - \sum_{t\in[a_i,d_i]} \pi_i^t(\{\theta_i\} \cup \widehat{\theta}_{I^{\le t}\setminus\{i\}}, \widehat{\theta}_{J^{\le t}}) \ge 0$;

(II) for any carrier j, $\sum_{t\in[a_j,d_j]} \omega_j^t\left(\widehat{\theta}_{I^{\le t}}, \{\theta_j\} \cup \widehat{\theta}_{J^{\le t}\setminus\{j\}}\right) - \sum_{t\in[a_j,d_j]} y_i^t\left(\widehat{\theta}_{I^{\le t}}, \{\theta_j\} \cup \widehat{\theta}_{J^{\le t}\{j\}}\right)c_j \ge 0$.

To measure an online mechanism's efficiency, we need to compare the social welfare realized by the mechanism with the *ex-post* maximum social welfare under complete information. If the platform knows the future auction dynamics and all of the agents bid truthfully, the *ex-post* maximum social welfare until period $t \in T$ can be found by solving the following integer program.

$$\boldsymbol{IP1}(I^{\le t}, J^{\le t}) : \max_{\left(x_i^\tau, y_j^\tau\right)} V_1(I^{\le t}, J^{\le t}) = \sum_{\tau\in[0,t]} \left(\sum_{i\in I^{\le t}} x_i^\tau v_i - \sum_{j\in J^{\le t}} y_j^\tau c_j \right) \tag{1}$$

$$s.t. \sum_{i\in\{i|l_i=l,i\in I^{\le t}\}} x_i^\tau = \sum_{j\in J^{\le t}} y_j^\tau q^l(b_j), \forall l \in L, \forall \tau \le t \tag{2}$$

$$\sum_{\tau\in[a_i,d_i]} x_i^\tau \le q_i, \forall i \in I^{\le t} \tag{3}$$

$$\sum_{\tau\in[a_j,d_j]} y_j^\tau \le 1, \forall j \in J^{\le t} \tag{4}$$

$$x_i^\tau \in Z_0^+, \forall \tau \le t, \forall i \in I^{\le t} \tag{5}$$

$$x_i^\tau = 0, \forall \tau \notin [a_i, d_i], \forall i \in I^{\le t} \tag{6}$$

$$y_j^\tau \in \{0,1\}, \forall \tau \le t, \forall j \in J^{\le t} \tag{7}$$

$$y_j^\tau = 0, \forall \tau \notin [a_j, d_j], \forall j \in J^{\le t} \tag{8}$$

Constraint (2) ensures supply and demand balance in each period t; constraints (3) and (4) mean that shippers cannot obtain more than their demands and carriers cannot sell more than their supplies, respectively.

An online bundle double auction mechanism is hardly efficient for the two following reasons. First, the platform cannot forecast the future dynamics. Second, as pointed out by Myerson and Satterthwaite (1983), no bilateral exchange can be simultaneously efficient, individually rational, budget-balanced, and incentive-compatible even in a static environment. In this paper we aim to design *feasible*, *incentive-compatible*, *individually rational*, and *budget-balanced* online bundle double auction mechanisms for transport service transactions with dynamic demands and supplies. In addition, we also strive for high mechanism efficiency.

4. The OBDA mechanism

In this section, we present the Online Bundle Double Auction (OBDA) mechanism for the basic case where each shipper wants to procure a unit of transport service on a lane and each carrier can supply a bundle of transport services (multiple units on multiple lanes).

4.1. The framework of the OBDA mechanism

The framework of the OBDA mechanism consists of four modules: the pre-determined pricing rule, admission rule, allocation and payment rules, and survival rule.

Pre-determined pricing rule. Given a set of shippers and a set of carriers, it sets a pre-buying price for each shipper's trading unit and a pre-selling price for each carrier's bundle. The admission, allocation, and payment decisions hereafter are based on it.

Admission rule. This rule determines whether an agent upon arrival is allowed to enter transactions.

Allocation and payment rules. The allocation rule determines whether a shipper can get the transport service and whether a carrier can sell its bundle of transport services during their stay periods. The payment rule determines the shippers' payments and the carriers' revenues.

Survival rule. This rule decides whether an agent has a chance to trade in the following periods. It can increase agents' opportunities to trade, thus improving the mechanism's efficiency. Meanwhile, it can prevent the agents from misrepresenting arrival times when they face trade failures due to supply–demand imbalance.

When a shipper/carrier arrives at the market and submits a bid/ask to the platform, it faces a series of decisions specified by the above rules. Before introducing this process, we need some new definitions. At the start of period t, the state of a shipper i with $\widehat{a}_i \le t \le \widehat{d}_i$ (a carrier j with $\widehat{a}_j \le t \le \widehat{d}_j$) is one of the following three ones: *traded*, *rejected*, or *active*. A *traded* agent indicates that it has gotten a transaction and its payment (revenue) has been determined. A *rejected* one indicates that it is rejected by the platform and cannot get a transaction. An *active* one is neither *rejected* nor *traded*, and whether or not to trade is undecided. Fig. 2 shows the framework of the OBDA mechanism.

We see from Fig. 2 that, upon the arrival of a new agent, an admission decision is first made to decide whether it can be admitted to active status. The agent is active if its bid (ask) price is higher (lower) than the admission price. Otherwise, the agent is rejected. Then, all the active agents compete for allocation. If an agent gets allocation, it can trade. If it fails and does not survive, it is rejected. Otherwise, the agent remains to be active in the next period. The process is repeated until its departure time. In the admission, allocation, payment, and survival decisions, the admission price and buying/selling price are determined by the pre-determined pricing rule.

4.2. The details of the OBDA mechanism

In this subsection we describe each module of the OBDA mechanism in detail.

Module I: Pre-determined pricing rule

For a given set I of shippers and a given set J of carriers, we pre-determine the price for each shipper $i \in I$ and carrier $j \in J$. Note that the set I (J) is not specific to certain shippers (carriers). For example, in the admission rule, I refers to $\overline{I}^t \cup \{i\}$ while calculating shipper i's admission price but it refers to $\overline{I}^t$ while calculating carriers' admission prices, where $\overline{I}^t$ is the set of shippers who are active in period t.

The pre-determined pricing rule for shipper $i \in I$ and carrier $j \in J$ is as follows:

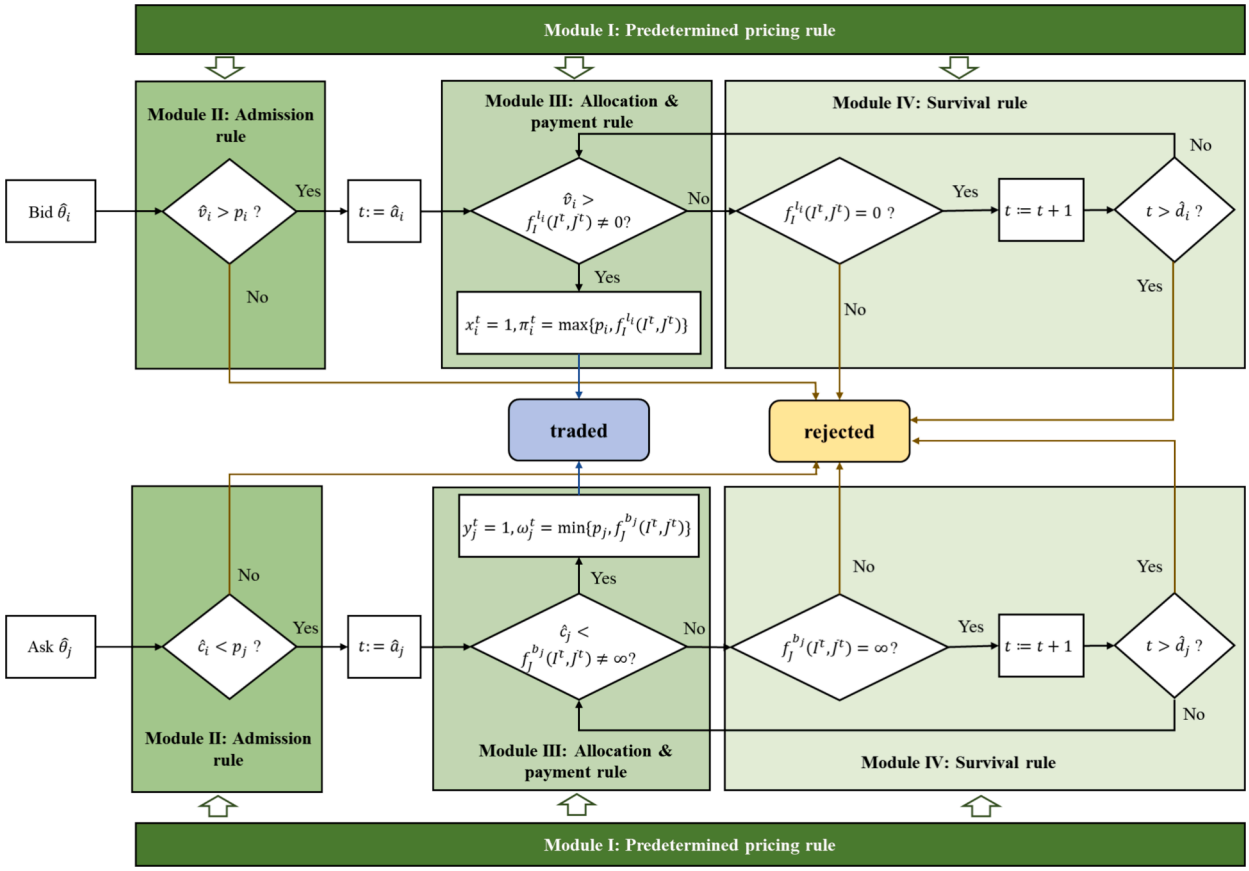


Fig. 2. The framework of the OBDA mechanism.

Step 1: Solve the integer program ***IP2***(I,J) and find the optimal solution $\{(\alpha_i^*)_{i\in I}, (\beta_j^*)_{j\in J})\}$.[4]

$$\textbf{\textit{IP2}}(I,J): \max_{(\alpha_i,\beta_j)} \widehat{V}_1(I,J) = \sum_{i\in I}\alpha_i\widehat{v}_i - \sum_{j\in J}\beta_j\widehat{c}_j \tag{9}$$

$$s.t. \sum_{i\in\{i|l_i=l,i\in I\}}\alpha_i = \sum_{j\in J}\beta_j q^l(b_j), \forall l\in L \tag{10}$$

$$\alpha_i \in \{0,1\}, \forall i\in I \tag{11}$$

$$\beta_j \in \{0,1\}, \forall j\in J \tag{12}$$

In objective function (9), α_i and β_j denote whether shipper i and carrier j trade, respectively. Constraint (10) guarantees supply and demand balance.

Step 2: Let $\widetilde{B} = \left\{b \middle| \sum_{j\in\{j|b_j=b,j\in J\}}\beta_j^* \geq 1, b\in B\right\}$ and $B^* = \left\{b \middle| \sum_{j\in\{j|b_j=b,j\in J\}}\beta_j^* \geq 2, b\in B\right\}$. The pre-selling price for carrier j who asks for bundle $b_j\in B\backslash B^*$ is set as $f_J^{b_j}(I,J) = \infty$.

Step 3: For each bundle $b\in B^*$, sort the ask prices of the carriers with $\beta_j^* = 1$ in a strictly increasing order. Let j_b^* be the carrier with the highest ask price for bundle b. For each carrier j with $b_j = b$, its pre-selling price is set as $f_J^{b_j}(I,J) = \widehat{c}_{j_b^*}$.

Step 4: Let $L^* = \left\{l \middle| \sum_{i\in\{i|l_i=l,i\in I\}}\alpha_i^* > \sum_{b\in\widetilde{B}}q^l(b), l\in L\right\}$. The pre-buying price for shipper i who bids for lane $l_i\in L\backslash L^*$ is set as $f_I^{l_i}(I,J) = 0$.

Step 5: For each lane $l\in L^*$, sort the bid prices of the shippers with $\alpha_i^* > 0$ in a strictly decreasing order. Let i_l^* be the shipper with the $\sum_{b\in\widetilde{B}}q^l(b)-$th lowest bid price for lane l. Then the pre-buying price for shipper i with $l_i = l$ is set as $f_I^{l_i}(I,J) = \widehat{v}_{i_l^*}$.

This pre-determined pricing rule (**Step 1**) begins with a transaction allocation between the shippers and carriers to maximize the social welfare when all the agents bid truthfully. **Steps 2–5** describe the pricing rule that can incentivize the agents to bid truthfully so that actual transactions do not deviate from the one specified by **Step 1**. Specifically, **Steps 2–3** determine each carrier's pre-selling price, where the price is set by the supply bundle. That is, for a given supply bundle, the pre-selling prices for all carriers are the same. According to the optimal solution to ***IP2***(I,J), the supply bundle set B can be divided into two subsets, $\widetilde{B}$ and $B\backslash\widetilde{B}$, which represent the bundles that are traded and not sold in the optimal allocation, respectively. For bundles $B\backslash\widetilde{B}$, although no carriers can trade in this period, they may have the opportunity to trade in some subsequent periods. So in **Step 2**, we consider keeping these carriers active by setting the pre-selling prices to be ∞.

For traded bundles $\widetilde{B}$, we use the Bundle Double Auction (BDA) mechanism proposed by Xu et al. (2016) to set the pre-selling prices. The BDA mechanism is a one-period double auction mechanism where each shipper wants one unit of the item, while each carrier supplies a bundle of items. Its idea is to remove the least profitable trade for each bundle from the optimal allocation and determine the buying price for each bundle as the highest bid price removed and the selling price for each item as the lowest ask price removed, respectively. Note that $\widetilde{B}$ can be further divided into two subsets B^* and $\widetilde{B}\backslash B^*$, which represent the bundles sold by at least two carriers and the bundles sold by only one carrier in the optimal allocation, respectively. For each bundle $b\in B^*$, we directly employ the BDA mechanism to set the pre-selling price as shown in **Step 3**. That is, the pre-selling price $f_J^b(I,J)$ is set to be the highest ask price of the carriers with positive allocations ($\beta_j^* \geq 0$) in the optimal solution to ***IP2***(I,J). However, for bundles $\widetilde{B}\backslash B^*$, if we directly apply the BDA mechanism, then all carriers asking for these bundles will be eliminated, thereby resulting in a loss of efficiency. So we use the same pricing rule as that for $B\backslash B^*$ (**Step 2**) to improve potential efficiency.

Similar to **Steps 2–3**, **Steps 4–5** determine each shipper's pre-buying price by the transport lane. For lanes $L\backslash L^*$, **Step 4** sets the pre-buying prices to 0 which is lower than any shipper's bid price. For lane $l\in L^*$, the pre-buying price is set based on the BDA mechanism as shown in **Step 5**.

According to the allocation rule described later, the carriers with ask prices less than their non-infinite pre-selling prices and the shippers with bid prices larger than their non-zero pre-buying prices can trade and the others will be eliminated. For these traded agents, the trading prices are independent of their ask (bid) prices. For those eliminated ones, they would get negative utility if they trade. So **Steps 2–5** can incentivize the agents to bid their prices truthfully. In addition, for each $b\in\widetilde{B}$, **Steps 2–3** remove one carrier with a positive allocation in the optimal allocation. For each lane $l\in L^*$, **Step 5** removes corresponding $\sum_{b\in\widetilde{B}}q^l(b)$ shippers with positive allocations in the optimal allocation. Given single-unit demand, the remaining demand and supply are balanced. In the optimal solution to ***IP2***(I,J), we have that $f_J^b(I,J) = \widehat{c}_{j_b^*} \leq \sum_{l\in L}q^l(b)\widehat{v}_{i_l^*} = \sum_{l\in L}q^l(b)f_I^l(I,J)$ for each bundle b. So the platform can earn a non-negative profit from each traded bundle, i.e., the pricing scheme generated by **Steps 2–5** is also budget-balanced.

[4] The perturbation method proposed by Babaioff and Walsh (2005) can be used to break ties and find the single optimal solution.

Module II: Admission rule

The admission price p_i for shipper i and price p_j for carrier j are defined as follows:

$$p_i = \begin{cases} 0, & \widehat{a}_i = 0 or \widehat{d}_i - \widehat{a}_i = K \\ \max\limits_{\tau \in [\max(0,\widehat{d}_i - K), \widehat{a}_i - 1]} \left\{ f_I^{l_i}(\overline{I}^{\tau} \cup \{i\}, \overline{J}^{\tau}) \right\}, & \widehat{a}_i > 0 and \widehat{d}_i - \widehat{a}_i < K \end{cases} \tag{13}$$

$$p_j = \begin{cases} \infty, & \widehat{a}_j = 0 or \widehat{d}_j - \widehat{a}_j = K \\ \min\limits_{\tau \in \left[\max\left(0,\widehat{d}_j - K\right), \widehat{a}_j - 1\right]} \left\{ f_J^{b_j}(\overline{I}^{\tau}, \overline{J}^{\tau} \cup \{j\}) \right\}, & \widehat{a}_j > 0 and \widehat{d}_j - \widehat{a}_j < K \end{cases} \tag{14}$$

$\overline{J}^t$ is the set of carriers whose asks are active in period t. If shipper i's bid price $\widehat{v}_i$ is higher than the admission price p_i or carrier j's ask price $\widehat{c}_j$ lower than the admission price p_j, it is admitted to be active. Otherwise, it is rejected.

According to Eq.(13), the admission prices for all shippers arriving in period 0 are set to 0, i.e., the shippers arriving in period 0 are active. This is because we do not have any transaction information prior to period 0, then the admission prices are initialized to 0. For shipper i arriving in other periods, there are two pricing cases according to her submission of $(\widehat{a}_i, \widehat{d}_i)$: (i) $\widehat{d}_i - \widehat{a}_i = K$, and (ii) $\widehat{d}_i - \widehat{a}_i < K$. Recall that K is the upper bound of the agents' patience, i.e., $d_i - a_i \le K$ for all i. If $\widehat{d}_i - \widehat{a}_i = K$, shipper i must submit its arrival time a_i and departure time d_i truthfully since a shipper cannot submit $\widehat{a}_i$ earlier than a_i and $\widehat{d}_i$ later than d_i. So we allow the shipper to proceed directly to the subsequent transaction by setting the admission prices as 0. If $\widehat{d}_i - \widehat{a}_i < K$, shipper i are likely to misrepresent its arrival time a_i and departure time d_i. To avoid these manipulations, the determination of the admission price p_i takes into account the pricing schemes from period $(\widehat{d}_i - K)$ to period $(\widehat{a}_i - 1)$. In addition, p_i is monotonically increasing over periods. So shipper i will get a higher p_i if it arrives late or leaves early. The determination (Eq.(14)) of the admission prices for the carriers has similar implications. In summary, this admission rule can overcome the difficulty in the design of online mechanisms where the agents may misrepresent their time information.

Module III: Allocation and payment rule

In period t, we first calculate the pre-buying price $f_I^{l_i}(\overline{I}^t, \overline{J}^t)$ for each shipper $i \in \overline{I}^t$ and the pre-selling price $f_J^{b_j}(\overline{I}^t, \overline{J}^t)$ for each carrier $j \in \overline{J}^t$. If $f_I^{l_i}(\overline{I}^t, \overline{J}^t) \neq 0$ and $\widehat{v}_i > f_I^{l_i}(\overline{I}^t, \overline{J}^t)$, $x_i^t = 1$, and shipper i can buy one unit of transport services on lane l_i in period t. Similarly, if $f_J^{b_j}(\overline{I}^t, \overline{J}^t) \neq \infty$ and $\widehat{c}_j < f_J^{b_j}(\overline{I}^t, \overline{J}^t)$, $y_j^t = 1$, and carrier j can sell a bundle of transport services b_j in period t. Otherwise, $x_i^t = 0$ and $y_j^t = 0$.

If shipper i trades in period t, its payment π_i^t is $\max\left\{p_i, f_I^{l_i}(\overline{I}^t, \overline{J}^t)\right\}$. If carrier j trades in period t, its payoff ω_j^t is $\min\left\{p_j, f_J^{b_j}(\overline{I}^t, \overline{J}^t)\right\}$. Otherwise, $\pi_i^t = 0$ and $\omega_j^t = 0$.

According to the pre-determined pricing rule, there is a balance between the total demand of the shippers with bid prices above the non-zero pre-buying prices and the total supply of the carriers with ask prices below the non-infinite pre-selling prices. So in the allocation rule, we allow these shippers and carriers to trade and eliminate the others. As for the payment rule, the payment (payoff) of a traded agent is structured similarly to the admission price setting as in Eqs.(13)-(14). So it can incentivize the agents to bid their arrival and departure times truthfully.

Module IV: Survival rule

If an active shipper does not trade in $t < \widehat{d}_i$ and its pre-buying price in period t is $f_I^{l_i}(\overline{I}^t, \overline{J}^t) = 0$, it remains active in period $t+1$ and its admission price remains unchanged. Similarly, if an active carrier does not trade in $t < \widehat{d}_j$ and its pre-selling price in period t is $f_J^{b_j}(\overline{I}^t, \overline{J}^t) = \infty$, it remains active in period $t+1$ and its admission price remains unchanged. Otherwise, the agent is rejected.

Recall that in the pre-determined pricing rule, some agents do not trade and their pre-buying (selling) prices are set to 0 (∞). So this survival rule retains these agents for the next period. On the one hand, it can increase agents' opportunities to trade in the future, thus improving the efficiency. On the other hand, it can prevent the agents from misrepresenting arrival times when they face trade failures due to supply–demand imbalance.

Remark 2 If an agent is rejected and its type does not change, it has no incentive to resubmit a new bid. Since bidding truthfully is its weak dominant strategy, it will still bid truthfully if the agent rebids. From Eqs.(13) and (14), the calculation of the admission price considers several periods before the arrival of the agent. Then the agent may get an admission price that is not lower (higher) than its valuation (cost) and thus it would still be rejected. An agent may rebid in one of the following three situations: (i) an agent's valuation/cost changes for some exogenous reasons; (ii) a carrier adjusts to ask for a different bundle; (iii) the shipper (carrier) converts to the carrier (shipper) status to bid (this is often discussed in some literature on carrier collaboration mechanisms, e.g., Li et al. (2015), Lai et al. (2017), and Lyu et al. (2019). In these situations, this agent's bid/ask can be considered as a new one.

4.3. Example 1

In this subsection, we use a simple example to illustrate how the OBDA mechanism works. We consider a small transport network consisting of two lanes m and n. Carriers can ask for three types of bundles ($1m$, $1n$, and $1m1n$), where $1m1n$ means that a carrier can

provide one truckload of transport service for both lanes m and n. Table 2 shows the bids from the shippers and the asks from the carriers in each period. We assume that $K = 2$.

By solving $\boldsymbol{IP1}\left(I^{\leq 2}, J^{\leq 2}\right)$, we can get the optimal allocation, i.e., shippers 1, 3, 5, 6 and carriers 3, 4 trade in period 0, shippers 7–9 and carriers 1, 2, 6 trade in period 1, shippers 2, 10–14 and carriers 7–10 trade in period 2. Then the *ex-post* maximum social welfare is $33.5\ (\underbrace{(8+6+7+6-7-11)}_{period\ 0} + \underbrace{(8+6.5+9-3.5-5-4)}_{period\ 1} + \underbrace{(7+7+6+8+7+6.5-6-7.5-3.5-11)}_{period\ 2} = 33.5)$.

Then we employ the OBDA mechanism to clear the market. In period 0, by Eqs. (13)-(14), the admission prices for all shippers (carriers) are 0 (∞), respectively. All the agents are active. Next we use the pre-determined pricing rule to determine each agent's pre-buying (pre-selling) price in period 0. Specifically, by solving $\boldsymbol{IP2}\left(\bar{I}^0, \bar{J}^0\right)$, all shippers and carriers trade in the optimal allocation. According to **Steps 3** and **5** of the pre-determined pricing rule, for bundle $1m$, j^*_{1m} is carrier 2; for bundle $1m1n$, j^*_{1m1n} is carrier 4; for lane m, i^*_m is shipper 3; and for lane n, i^*_n is shipper 6. Then we have that the pre-buying price $f^m_I\left(\bar{I}^0, \bar{J}^0\right)$ for shipper i with $l_i = m$ is 6, $f^n_I\left(\bar{I}^0, \bar{J}^0\right)$ for shipper i with $l_i = n$ is 6, the pre-selling price $f^{1m}_J\left(\bar{I}^0, \bar{J}^0\right)$ for carrier j with $b_j = 1m$ is 5, and $f^{1m1n}_J\left(\bar{I}^0, \bar{J}^0\right)$ for carrier j with $b_j = 1m1n$ is 11. According to the allocation rule, the bid prices of shippers 1, 2, and 5 are higher than their pre-buying prices and the ask prices of carriers 1 and 3 are lower than their pre-selling prices, so they can trade in this period. Other agents (shippers 3, 4, 6, and carriers 2 and 4) are rejected. According to the payment rule, the payment of shipper 1 is $\max\left\{p_i = 0, f^m_I\left(\bar{I}^0, \bar{J}^0\right) = 6\right\} = 6$. Similarly, shippers 2 and 5 pay 6 for their demand, respectively and carriers 1 and 3 get 5 and 11 for their supply, respectively. Since there are no shippers with a pre-buying price of 0 and no carriers with a pre-selling price of ∞, no bids and asks survive to the next period.

In period 1, we first calculate the admission prices for the newly arrived agents. For shipper 7, $\widehat{d}_7 - \widehat{a}_7 = 2 - 1 < K = 2$ and then $p_7 = \max_{\tau \in [\max(0, \dot{d}_i - K), \dot{a}_i - 1]} \left\{f^m_I(\bar{I}^\tau \cup \{shipper\ 7\}, \bar{J}^\tau)\right\} = f^m_I\left(\bar{I}^0 \cup \{shipper\ 7\}, \bar{J}^0\right)$ from Eq.(13). By solving $\boldsymbol{IP2}\left(\bar{I}^0 \cup \{shipper\ 7\}, \bar{J}^0\right)$, shippers 1–3, 5–7, and carriers 1–4 trade in the optimal allocation. According to **Step 5** of the pre-determined pricing rule, for lane m, i^*_m is shipper 2 and then shipper 7's admission price is $p_7 = f^m_I\left(\bar{I}^0 \cup \{shipper\ 7\}, \bar{J}^0\right) = \widehat{v}_2 = 7$. Similarly, from (13)-(14), the admission prices for shippers 8–9 are 0 and for carriers 5–8 are ∞, respectively. Then, shippers 7–9 and carriers 5–8 are active. From the pre-determined pricing rule, shippers 7-9's buying prices are 0, and carriers 5-8's selling prices are ∞. So they do not trade in this period but remain active in the next period.

In period 2, the agent set contains the newly arrived agents (shippers 10–14 and carriers 9–10) and the previous surviving agents (shippers 7–9 and carriers 5–8). The admission prices for shippers 10–14 are 0, 0, 0, 0, and 0, and for carriers 9–10 are ∞ and ∞, respectively. All the agents are active in period 2. From the pre-determined pricing rule, $f^m_I\left(\bar{I}^2, \bar{J}^2\right) = 6.5$, $f^n_I\left(\bar{I}^2, \bar{J}^2\right) = 7$, $f^{1m}_J\left(\bar{I}^2, \bar{J}^2\right) = 5.5$, $f^{1n}_J\left(\bar{I}^2, \bar{J}^2\right) = 6$, and $f^{1m1n}_J\left(\bar{I}^2, \bar{J}^2\right) = 11$. Then shippers 8, 11, 13, and 14, and carriers 5, 7, and 10 are rejected. Shippers 7, 9, 10, and 12 pay 7, 7, 6.5, and 7 for their demand, respectively. Carriers 6, 8, and 9 get 6, 5.5, and 11 for their supply, respectively.

By the end of period 2, seven shippers among the 14 shippers and five carriers among the ten carriers have successfully traded. The shippers' total utility is $8.5\ (\underbrace{(8+7+7-6\times 2-6)}_{period\ 0} + \underbrace{(8+7+9+8-6.5-7-7\times 2)}_{period\ 2} = 8.5)$ and the carriers' total utility is $13\ (\underbrace{(5+11-3.5-7)}_{period\ 0} + \underbrace{(5.5+6+11-3.5-4-7.5)}_{period\ 2} = 13)$. The platform exactly matches the supply and demand in every period and obtains a profit of $7\ (\underbrace{(6\times 3-5-11)}_{period\ 0} + \underbrace{(6.5+7\times 3-5.5-6-11)}_{period\ 2} = 7)$. The OBDA mechanism yields the social welfare of $28.5\ (\underbrace{(8+7+7-3.5-7)}_{period\ 0} + \underbrace{(8+7+9+8-3.5-4-7.5)}_{period\ 2} = 28.5)$ and realizes the efficiency of 85.07 % (28.5/33.5 = 85.07%).

4.4. The properties

In this subsection we show that the OBDA mechanism has good properties.

Theorem 1 *The OBDA mechanism is strongly feasible, ex-post weakly budget-balanced, and ex-post individually rational.*

Proof. Please see Appendix A.1.

Theorem 1 shows that the OBDA mechanism can exactly match the demand and supply, and ensure a non-negative profit for the platform. Moreover, each agent can get a non-negative utility from participating in the auction. Hence, the OBDA mechanism is valid for transport service transactions with dynamic demands and supplies. It can incentivize the platform to provide exchange services, and can also attract and retain agents to participate in platform transactions.

Theorem 2. *The OBDA mechanism is incentive-compatible.*

Proof. Please see Appendix A.2.

Theorem 2 shows that the agents will bid their valuations and costs truthfully in equilibrium regardless of the others' bidding strategies and future dynamics. Hence, the problem of computing the optimal bids reduces to the problem of estimating the marginal costs and valuations for the transport. In addition, the agents would immediately join the market and leave until they lose their patience. In practice, this can simplify the agents' bidding strategies, i.e., bidding truthfully, and attract more agents to engage in online platform transactions.

We now evaluate the efficiency loss of the OBDA mechanism.

Theorem 3 *If $d_i - a_i = K$ for any shipper i and $d_j - a_j = K$ for any carrier j, the efficiency loss of the OBDA mechanism is bounded by*

Table 2
Bids and asks from the agents in each period.

Period	Shipper:$\widehat{\theta}_i = \{\widehat{a}_i, \widehat{d}_i, q_i, \widehat{v}_i\}$		Carrier:$\widehat{\theta}_j = \{\widehat{a}_j, \widehat{d}_j, \widehat{c}_j\}$		
	m	n	$1m$	$1n$	$1m1n$
0	1:{0,1,1,8} 2:{0,2,1,7} 3:{0,0,1,6} 4:{0,2,1,5.5}	5:{0,0,1,7} 6:{0,1,1,6}	1:{0,1,3.5} 2:{0,2,5}		3:{0,1,7} 4:{0,0,11}
1	7:{1,2,1,8} 8:{1,3,1,6.5}	9:{1,3,1,9}	5:{1,3,5.5}	6:{1,3,4} 7:{1,2,6}	8:{1,3,7.5}
2	10:{2,4,1,7} 11:{2,3,1,6}	12:{2,4,1,8} 13:{2,3,1,7} 14:{2,4,1,6.5}	9:{2,3,3.5}		10:{2,4,11}

$O\left(\max\left\{\frac{1}{\min_{l\in L}[K^l]}, \frac{1}{\min_{b\in B}[K^b(K^b-1)]}\right\}\right)$, *where* K^l *and* K^b *are the expected numbers of optimal trading shippers for lane* l *and optimal trading carriers for bundle* b *in the efficient allocation in a period, respectively.*

Proof. Please see Appendix A.3.

Theorem 3 states that the efficiency loss of the OBDA mechanism is bounded and decreases with the number of trading agents in each period. The platform can improve efficiency by attracting more agents to participate in the market.

5. Two extensions

In this section, we consider two extensions where the shippers want to buy multi-unit transport services on one lane or multi-unit indivisible demand on multiple lanes. We modify the OBDA mechanism and propose the OBDA-padding mechanism for the former and the OBDA-SMP mechanism for the latter.

5.1. The case with multi-unit demand on one lane

In this subsection, we consider the case where shipper i wants to buy $q_i \geq 1$ units of transport services on lane l_i and the demands can be met by more than one carrier. In this case, the OBDA mechanism has the following problems: (i) the allocation is not valid, (ii) shippers will not bid truthfully, and (iii) the budget is unbalanced. The following example for one period of the transaction shows these.

Example 2 Table 3 lists the agent's bids.

From the pre-determined pricing rule in the OBDA mechanism, i_m^* will be shipper 4 and $f_I^m(I,J) = \widehat{v}_4 = 5$. Then there are five units of demand but six units of supply for lane m, which results in a valid transaction. If we allow shipper 4 to trade one unit, it has an incentive to lower the bid price so that the buying price is lower. If we set i_m^* to be shipper 5, $f_I^m(I,J) = \widehat{v}_5 = 3.5$, the budget is unbalanced ($2 \times f_I^m(I,J) + f_I^n(I,J) = 2\widehat{v}_5 + \widehat{v}_9 = 12 < 13 = \widehat{c}_4 = f_J^{2m1n}(I,J)$) even though shipper 4 bids truthfully. The example ends here.

The main cause of these problems is that in the case with multi-unit demand, the pre-determined pricing rule may set the buying price as the bid price of a shipper with under-satisfied demand, who can manipulate its bid to influence the buying price. In order for the pre-determined pricing rule to be viable, we need to ensure that i_l^*'s demand must be completely removed and meanwhile the supply cannot exceed the demand since the supply bundle is indivisible. To overcome these difficulties, we adopt the "supply-side padding" method (Guo, 2024b) to modify the pre-determined pricing rule in the OBDA mechanism and propose the OBDA-padding mechanism for the case with multi-unit demand on one lane.

The traditional "padding" method is proposed by Chu (2009). Guo et al. (2024) extended the traditional "padding" method to the multi-unit case with sellers' variable marginal costs and called it the "supply-side padding" method. The method first defines a supply-side padding, i.e., a phantom seller with a specific supply and a zero cost. Then, it makes an initial allocation with the padding to remove some sellers. Finally, it matches the remaining sellers and the original buyers. Each trading agent will pay or get the VCG

Table 3
The agents' bids.

Shipper:$\widehat{\theta}_i = \{q_i, \widehat{v}_i\}$		Carrier:$\widehat{\theta}_j = \{\widehat{c}_j\}$
m	n	$2m1n$
1:{1,8}	6:{1,8}	1:{9}
2:{2,7}	7:{1,7}	2:{10}
3:{2,6}	8:{1,6}	3:{11}
4:{2,5}	9:{1,5}	4:{13}
5:{1,3.5}		

payment (revenue). Note that this static method cannot directly be used to construct our pre-determined pricing rule for the two following reasons. First, Guo et al. (2024) only consider single-item case. Second, the VCG payment rule will generate differential buying prices for the trading shippers with different demands. Then the shippers may strategically delay arrivals to trade different quantities and thus gain a price advantage in the dynamic trading environment. Nevertheless, we can modify the original "padding" method to construct a new pre-determined pricing rule for our case. We first introduce a supply-side padding vector to convert the original auction into another one where the total demand exceeds total the supply. Then the uniform buying price is determined based on the BDA mechanism rather than the VCG payment rule as in the traditional "padding" method. Under this new rule, even if the total demand is greater than the total supply for a particular lane while allocating the transactions, we can still ensure the feasible allocation because the demand is divisible. The budget can be also balanced as long as the padding is large enough. Specifically, we state the pre-determined pricing rule as follows.

Module I: The modified pre-determined pricing rule

Step 1′: Determine the padding vector $\boldsymbol{S} = \{S_l | l \in L\}$ where $S_l = \max\{q_i | l_i = l, i \in I\}$.[5]

Step 2′: Solve the integer program $\boldsymbol{IP3}(I, J, \boldsymbol{S})$ and find the optimal solution $\{(\alpha_i')_{i\in I}, (\beta_j')_{j\in J}\}$, which maximizes the social welfare $\widehat{V}_2(I,J)$.

$$\boldsymbol{IP3}(I,J,\boldsymbol{S}): \max_{(\alpha_i,\beta_j)} \widehat{V}_2(I,J) = \sum_{i\in I}\alpha_i\widehat{v}_i - \sum_{j\in J}\beta_j\widehat{c}_j \tag{15}$$

$$s.t. \sum_{i\in\{i|l_i=l,i\in I\}}\alpha_i = \sum_{j\in J}\beta_j q^l(b_j) + S_l, \forall l \in L \tag{16}$$

$$\alpha_i \le q_i, \forall i \in I \tag{17}$$

$$\alpha_i \in Z_0^+, \forall i \in I \tag{18}$$

$$\beta_j \in \{0,1\}, \forall j \in J$$

In objective function (15), α_i denotes the number of trade units of shipper i. Constraint (16) guarantees supply and demand balance with the padding. Constraint (17) means that the shippers cannot obtain more than their demands. Constraint (18) means that α_i is an integer variable.

Step 3′: Let $\widetilde{B} = \left\{b \middle| \sum_{j\in\{j|b_j=b,j\in J\}}\beta_j' \ge 1, b \in B\right\}$ and $B^* = \left\{b \middle| \sum_{j\in\{j|b_j=b,j\in J\}}\beta_j' \ge 2, b \in B\right\}$. The pre-selling price for carrier j who asks for bundle $b_j \in B\backslash B^*$ is set as $f_J^{b_j}(I,J) = \infty$.

Step 4′: For each bundle $b \in B^*$, sort the ask prices of the carriers with $\beta_j^* = 1$ in a strictly increasing order. Let j_b^* be the carrier with the highest ask price for bundle b. For each carrier j with $b_j = b$, its pre-selling price, given I and J, is determined as $f_J^{b_j}(I,J) = \widehat{c}_{j_b^*}$.

Step 5′: Let $L^* = \left\{l \middle| \sum_{i\in\{i|l_i=l,i\in I\}}\alpha_i' > \sum_{b\in\widetilde{B}}q^l(b) + S_l, l \in L\right\}$. The pre-buying price for shipper i who bids for lane $l_i \in L\backslash L^*$ is set as $f_I^{l_i}(I,J) = 0$.

Step 6′: For each lane $l \in L$, sort the bid prices of the shippers with $\alpha_i' > 0$ in a strictly decreasing order. Label $\sum_{b\in\widetilde{B}}q^l(b)$ units with the lowest bid prices. Let i_l^* be the shipper with the highest bid price among them. Then the pre-buying price for shipper i with $l_i = l$, given I and J, is $f_I^{l_i}(I,J) = \widehat{v}_{i_l^*}$.

The modified pre-determined pricing rule is similar to the original one but incorporates the impacts of padding on the initial transaction allocation (Step 2′) and pre-buying price setting for shippers (Steps 5′-6′). For the sake of clarity, we re-analyse Example 2 to show the sense of the modified rule. Under the new rule, the padding vector $\boldsymbol{S} = \{S_m = 2, S_n = 1\}$. By solving $\boldsymbol{IP3}(I,J,\boldsymbol{S})$, all shippers and carriers 1–3 can trade in the optimal allocation. According to **Steps 4′** and **6′,** for lane m, i_m^* is shipper 4; for lane n, i_n^* is shipper 9; and for bundle $2m1n$, j_{2m1n}^* is carrier 3. Then we have that $f_I^m(I,J) = \widehat{v}_4 = 5$, $f_I^n(I,J) = \widehat{v}_9 = 5$, and $f_I^{2m1n}(I,J) = \widehat{c}_3 = 11$. For the shippers with bid prices above the pre-buying prices and the carriers with ask prices below the pre-selling price, there is more remaining demand than supply (five units of demand and four units of supply for lane m, and three units of demand and two units of supply for lane n). We can allocate these transactions among shippers 1–3 and 6–7 according to a bid price-independent attribute, e.g., bidding time. Then the mechanism can yield a valid allocation and a balanced budget ($2 \times f_I^m(I,J) + f_I^n(I,J) = 2\widehat{v}_4 + \widehat{v}_9 = 15 > 11 = \widehat{c}_3 = f_J^{2m1n}(I,J)$), and the shippers cannot obtain higher utilities by misrepresenting their bids either.

Accordingly, the allocation, payment, and survival rules for the shippers in the *OBDA-padding mechanism* should be modified for this multi-unit demand case as follows.

[5] The padding vector is determined in period 0 and remain constant in later periods.

Module III: The modified allocation and payment rules for the shippers

In period t, the buying price $f_I^{l_i}(\vec{I}^t,\vec{J}^t)$ for shipper $i \in \vec{I}^t$ is first calculated. Let $I_l^t = \left\{ i \middle| l_i = l, \widehat{v}_i > f_I^{l_i}(\vec{I}^t,\vec{J}^t) \neq 0, i \in \vec{I}^t \right\}$, and bt_i be shipper i's bidding time.[6] For lane l, sort the bidding time bt_i ($i \in I_l^t$) in increasing order. Without loss of generality, we assume that $bt_1 < bt_2 < .... < bt_{|I_l^t|}$. Let $J^t = \left\{ j \middle| \widehat{c}_j < f_J^{b_j}(\vec{I}^t,\vec{J}^t) \neq \infty, i \in \vec{J}^t \right\}$. Shipper $i \in I_l^t$ can obtain x_i^t units of transport services on lane l in period t, which is calculated as follows:

$$x_i^t = \begin{cases} \min\left\{ q_i, \sum_{j\in J^t} y_j^t q^l(b_j) - \sum_{n=1}^{i-1} x_n^t \right\}, & 2 \le i \le |I_l^t| \\ \min\left\{ q_i, \sum_{j\in J^t} y_j^t q^l(b_j) \right\}, & i = 1 \end{cases} \tag{19}$$

If shipper i trades in period t, its payment π_i^t is $x_i^t \max\left\{ p_i, f_I^{l_i}(\vec{I}^t,\vec{J}^t) \right\}$. Otherwise, $\pi_i^t = 0$.

Due to the padding introduced in the modified pre-determined pricing rule, the total demand of the shippers with bid prices above the non-zero pre-buying prices may be more than the total supply of the carriers with ask prices below the non-infinite pre-selling prices. So the modified allocation rule allocates the transactions among the shippers according to their bidding times. The shippers' demands are met in accordance with the order of their bidding times. Thus the shippers cannot gain an allocative advantage by manipulating their bid prices.

Module IV: The modified survival rule for shippers

If an active shipper does not trade in $t < \widehat{d}_i$ and its buying price in period t is $f_I^{l_i}(\vec{I}^t,\vec{J}^t) = 0$, it remains active in period $t+1$ and its admission price remains unchanged. If an active shipper i trades partially in $t < \widehat{d}_i$, then its bid is updated to $\widehat{\theta}_i = \left\{ \widehat{a}_i, \widehat{d}_i, l_i, q_i - \sum_{\tau=\widehat{a}_i}^{t} x_i^\tau, \widehat{v}_i \right\}$. It remains active in period $t+1$ and the admission price is updated to $\max\left\{ p_i, f_I^{l_i}(\vec{I}^t,\vec{J}^t) \right\}$. Otherwise, the shipper is rejected.

In the modified survival rule, in addition to the shippers with pre-buying prices of 0, the shippers whose demands are partially met are also retained for the next period. This can increase shippers' opportunities to trade in the case with multi-unit demand, thus improving the efficiency.

Now we use a simple example to illustrate the implementation of the OBDA-padding mechanism.

Example 3 Table 4 shows the bids and asks from agents in each period, in which shippers are ranked according to the bidding time. Some shippers want to procure multiple units of transport services. We set $K = 2$.

By solving $\boldsymbol{IP1}\left(I^{\le 2}, J^{\le 2}\right)$, we can get the optimal allocation, i.e., shippers 1 and 5 each buy two units and carriers 3 and 4 sell their supplies in period 0; shippers 7 and 9 buy two units and one unit respectively, and carriers 1, 5, and 7 sell their supplies in period 1; shippers 8–11 buy two, one, two and two units respectively, and carriers 2, 6, 8–10 sell their supplies in period 2. Then the ex-post maximum social welfare is $37.5 \ (\underbrace{(2(8+7)-7-11)}_{period\ 0} + \underbrace{(7.5\times 2+9-3.5-5.5-6)}_{period\ 1} + \underbrace{(2(6.5+6+7)+9-4.5-5-7.5-3.5-11)}_{period\ 2} = 37.5)$.

Then we employ the OBDA-padding mechanism to clear the market. First we determine the padding vector $\boldsymbol{S}$ as $S_m = 2$ as $S_n = 2$.

In period 0, from (13)-(14), the admission prices for all shippers are 0 and for all carriers are ∞, respectively. All the agents are active. Next we use the pre-determined pricing rule to determine each agent's pre-buying (pre-selling) price in period 0. Specifically, by solving $\boldsymbol{IP3}\left(\vec{I}^0, \vec{J}^0, \boldsymbol{S}\right)$, all shippers except shipper 4 buy all of their demands and all carriers sell their supplies in the optimal allocation. Shipper 4 buys one unit. According to **Steps 4′** and **6′** of the modified pre-determined pricing rule, for bundle $1m$, j_{1m}^* is carrier 2; for bundle $1m1n$, j_{1m1n}^* is carrier 4; for lane m, i_m^* is shipper 2; and for lane n, i_n^* is shipper 6. Then we have that the pre-buying price $f_I^m(\vec{I}^0,\vec{J}^0)$ for shipper i with $l_i = m$ is 5.5, $f_I^n(\vec{I}^0,\vec{J}^0)$ for shipper i with $l_i = n$ is 6.5, the selling price $f_J^{1m}(\vec{I}^0,\vec{J}^0)$ for carrier j with $b_j = 1m$ is 4.5, and $f_J^{1m1n}(\vec{I}^0,\vec{J}^0)$ for carrier j with $b_j = 1m1n$ is 11. According to the allocation rule, the bid prices of shippers 1, 3, and 5 are higher than their pre-buying prices and the ask prices of carriers 1 and 3 are lower than their pre-selling prices, so the other agents (shippers 2, 4, and carriers 2, 4) are rejected. According to the bidding times, shipper 1 pays 11 for two units of transport services on lane m, and shipper 5 pays 6.5 for one unit of transport service on lane n. Carriers 1 and 3 get 4.5 and 11 for their supplies, respectively. Shipper 3's bid price is larger than the pre-buying price and its demand is not met, so it remains active. Its bid remains unchanged and the admission price is updated to 5.5.

In period 1, we first calculate the admission prices for the newly arrived agents. For shipper 7, $\widehat{d}_7 - \widehat{a}_7 = 2 - 1 < K = 2$ and then $p_7 = \max_{\tau\in[\max(0,\widehat{d}_i-K),\widehat{a}_i-1]}\left\{ f_I^m(\vec{I}^\tau \cup \{shipper\ 7\}, \vec{J}^\tau) \right\} = f_I^m(\vec{I}^0 \cup \{shipper\ 7\}, \vec{J}^0)$ by Eq.(13). By solving $\boldsymbol{IP3}(\vec{I}^0 \cup \{shipper\ 7\}, \vec{J}^0, \boldsymbol{S})$, ship-

[6] Bidding time can be automatically recorded when the agent bids. It is more precise than the arrival time. For example, if the platform sets 9:00–10:00 each day as period 0 for clearing, and when a shipper bids at 9:05, then the bidding time is 9:05 and the arrival time is period 0 and.

Table 4
Bids and asks from the agents in each period.

Period	Shipper:$\widehat{\theta}_i = \{\widehat{a}_i, \widehat{d}_i, q_i, \widehat{v}_i\}$			Carrier:$\widehat{\theta}_j = \{\widehat{a}_j, \widehat{d}_j, \widehat{c}_j\}$	
	m	n	$1m$	$1n$	$1m1n$
0	1:{0,1,2,8} 2:{0,1,2,5.5} 3:{0,1,1,6} 4:{0,0,2,5}	5:{0,0,2,7} 6:{0,1,2,6.5}	1:{0,1,3.5} 2:{0,1,4.5}		3:{0,1,7} 4:{0,0,11}
1	7:{1,2,2,7.5} 8:{1,3,2,6.5}	9:{1,3,2,9}	5:{1,3,5.5}	6:{1,3,5} 7:{1,2,6}	8:{1,3,7.5}
2	10:{2,3,2,6}	11:{2,3,2,7} 12:{2,4,2,6.5}	9:{2,3,3.5}		10:{2,4,11}

pers 1–3, and 7 buy two, one, one, and two units respectively, and all carriers sell their supplies in the optimal allocation. According to **Step 6′** of the modified pre-determined pricing rule, for lane m, i_m^* is shipper 3. Then shipper 7′s admission price is $p_7 = f_I^m(\bar{I}^0 \cup \{shipper\ 7\}, \bar{J}^0) = \widehat{v}_3 = 6$. Similarly, by Eqs.(13)-(14), the admission prices for shippers 8–9 are 0 and for carriers 5–8 are ∞, respectively. Then, all the agents are active. From the pre-determined pricing rule, all the shippers' pre-buying prices are 0 and all the carriers' pre-selling prices are ∞ in this period. So they do not trade in this period but will remain active in the next period except for shipper 2, who will leave the market after period 1.

In period 2, the agent set contains the newly arrived agents (shippers 10–12 and carriers 9–10) and the previous surviving agents (shippers 7–9 and carriers 5–8). The admission prices are 0, 0, and 0 for shippers 10–12, and 5.5 and ∞ for carriers 9–10, respectively. All the agents are active in period 2. According to the pre-determined pricing rule, $f_I^m(\bar{I}^2, \bar{J}^2) = 6$, $f_I^n(\bar{I}^2, \bar{J}^2) = 6.5$, $f_J^{1m}(\bar{I}^2, \bar{J}^2) = 5.5$, $f_J^{1n}(\bar{I}^2, \bar{J}^2) = 6$, and $f_J^{1m1n}(\bar{I}^2, \bar{J}^2) = 11$. According to the bidding times, shippers 7 and 9 pay 12 and 13 for two units of transport services, respectively. Carriers 6, 8, and 9 get 6, 5.5, and 11 for their bundles, respectively.

By the end of period 2, the shippers' total utility is $13.5 = \underbrace{(8 \times 2 + 7 - 5.5 \times 2 - 6.5)}_{period\ 0} + \underbrace{(7.5 \times 2 + 9 \times 2 - 6 \times 2 - 6.5 \times 2)}_{period\ 2} = 13.5$) and the carriers' total utility is $11.5 = \underbrace{(4.5 + 11 - 3.5 - 7)}_{period\ 0} + \underbrace{(5.5 + 6 + 11 - 3.5 - 5 - 7.5)}_{period\ 2} = 11.5$). The platform achieves an exact match between the supply and demand in every period and obtains a profit of $4.5 = \underbrace{(5.5 \times 2 + 6.5 - 4.5 - 11)}_{period\ 0} + \underbrace{(6 \times 2 + 6.5 \times 2 - 5.5 - 6 - 11)}_{period\ 2} = 4.5$). The OBDA-padding mechanism yields the social welfare of $29.5 = \underbrace{(8 \times 2 + 7 - 3.5 - 7)}_{period\ 0} + \underbrace{(7.5 \times 2 + 9 \times 2 - 3.5 - 5 - 7.5)}_{period\ 2} = 29.5$) and realizes the efficiency of 78.67 % (29.5/37.5 = 78.67%). The example ends here.

Next we analyze the properties of the OBDA-padding mechanism.

Theorem 4 T*he OBDA-padding mechanism is strongly feasible, ex-post weakly budget-balanced, and ex-post individually rational.*

Proof. Please see Appendix A.4.

Theorem 5 T*he OBDA-padding mechanism is incentive-compatible.*

Proof. Please see Appendix A.5.

Theorems 4 and **5** show that the OBDA-padding mechanism has good properties even if the platform allows the shippers to bid multiple units of demand.

Theorem 6 shows the efficiency loss bound of the OBDA-padding mechanism. Suppose that for each lane $l \in L$, the shippers' demands are drawn independently and identically from a positive discrete distribution with an expectation of $\bar{q}^l$.

Theorem 6 T*he efficiency loss of the OBDA-padding mechanism is bounded by* $O\left(\max\left\{\max_{l\in L}\left[\frac{S_l + \sum_{b\in B} q^l(b)}{K^l \bar{q}^l}\right], \max_{b\in B}\left[\frac{1}{K^b(K^b-1)}\right]\right\}\right)$, *if* $d_i - a_i = K$ *for any shipper i and* $d_j - a_j = K$ *for any carrier j.*

Proof. Please see Appendix A.6.

Theorem 6 states that the efficiency loss of the OBDA-padding mechanism is bounded for a fixed padding and decreases with the number of trading agents. The OBDA-padding mechanism has less efficiency than the OBDA mechanism because of the following two reasons. First, the padding introduced in the OBDA-padding mechanism may remove more profitable transactions. Second, the OBDA-padding mechanism allocates transactions according to bidding times.

5.2. The case with multi-unit indivisible demand on multiple lanes

In this subsection, we further extend the results to the case where a shipper wants to procure multi-unit indivisible transport

services on multiple lanes (a bundle of demand) from only one carrier.[7] Let shipper i's bid be $\widehat{\theta}_i = \{\widehat{a}_i, \widehat{d}_i, o_i, \widehat{v}_i(o_i)\}$, where $o_i \in O$ is its demand bundle and O is the set of possible demand bundles, and $\widehat{v}_i(o_i)$ is the bid price for the demand bundle o_i. In this case, the *ex-post* maximum social welfare until period $t \in T$ can be found by solving the following integer program.

$$\boldsymbol{IP4}(I^{\leq t}, J^{\leq t}) : \max_{\left(x_i^\tau, y_j^\tau, z_{ij}^\tau\right)} V_2(I^{\leq t}, J^{\leq t}) = \sum_{\tau \in [0,t]} \left[\sum_{i \in I^{\leq t}} x_i^\tau v_i(o_i) - \sum_{j \in J^{\leq t}} y_j^\tau c_j \right] \tag{20}$$

$$s.t. \sum_{i \in I^{\leq t}} z_{ij}^\tau q^l(o_i) = y_j^\tau q^l\left(b_j\right), \forall j \in J^{\leq t}, \forall l \in L, \forall \tau \leq t \tag{21}$$

$$\sum_{j \in J^{\leq t}} z_{ij}^\tau = x_i^\tau, \forall i \in I^{\leq t}, \forall \tau \leq t \tag{22}$$

$$\sum_{\tau \in [a_i, d_i]} x_i^\tau \leq 1, \forall i \in I^{\leq t} \tag{23}$$

$$\sum_{\tau \in \left[a_j, d_j\right]} y_j^\tau \leq 1, \forall j \in J^{\leq t} \tag{24}$$

$$x_i^\tau \in \{0,1\}, \forall \tau \leq t, \forall i \in I^{\leq t} \tag{25}$$

$$x_i^\tau = 0, \forall \tau \notin [a_i, d_i], \forall i \in I^{\leq t} \tag{26}$$

$$y_j^\tau \in \{0,1\}, \forall \tau \leq t, \forall j \in J^{\leq t} \tag{27}$$

$$y_j^\tau = 0, \forall \tau \notin \left[a_j, d_j\right], \forall j \in J^{\leq t} \tag{28}$$

$$z_{ij}^\tau \in \{0,1\}, \forall \tau \leq t, \forall i \in I^{\leq t}, \forall j \in J^{\leq t} \tag{29}$$

In objective function (20), z_{ij}^τ denotes whether shipper i procures the transport service from carrier j in period τ. Constraints (21)-(22) balance the demand and supply in each period. Constraints (23)-(24) mean that each shipper cannot obtain more than its bundle and each carrier cannot sell more than its bundle, respectively.

In this case, the definition of strong feasibility needs to be adjusted as follows but the definitions of the other properties remain unchanged.

Definition 5 (Strong feasibility). *An online bundle double mechanism $M = \{(x, y), (\pi, \omega)\}$ is strongly feasible in the case with multi-unit indivisible demand on multiple lane if $\sum_{i \in I^{\leq t}} z_{ij}^\tau q^l(o_i) = \sum_{j \in J^{\leq t}} y_j^t q^l\left(b_j\right)$, $\forall l \in L, \forall t \in T$*

Note that the OBDA and OBDA-padding mechanisms cannot be applied to this case directly. First, they cannot ensure a valid allocation that a shipper's demand is fully allocated to one carrier. Second, it is difficult to ensure that the agents tell the truth because one supply bundle can match different sets of demand bundles. For example, if $O = \{1m, 2m, 1n, 2m1n\}$, then the supply bundle $2m1n$ can match the set of demands $\{1m, 1m, 1n\}$, the set $\{2m, 1n\}$, or the set $\{2m1n\}$. From the pre-determined pricing rule of the OBDA mechanism, if a supply for bundle $2m1n$ is removed, there would be three options for the removal of demand: (i) two lowest-utility demands for bundle $1m$ and one lowest-utility demand for bundle $1n$, (ii) one lowest-utility demand for bundle $2m$ and one lowest-utility demand for bundle $1n$, and (iii) one lowest-utility demand for $2m1n$. Different removal options correspond to different pricing schemes for the shippers, which induce the shippers to misrepresent their prices.

To overcome this difficulty, we modify the pre-determined pricing rule of the OBDA mechanism by pre-determining a *Single Matching Pattern (SMP)*, whereby a supply bundle can only be matched to a given set of demands. We called it the OBDA-SMP mechanism. Let B_e be the set of bundles, each of which is associated with a fixed set of demands, where $e \in E$ and E is the set of matching patterns. Consider the example with $O = \{1m, \quad 2m, \quad 1n, \quad 2m1n\}$ and $B = \{1m, \quad 2m1n\}$. Three matching patterns are available, i.e., $|E| = 3$, and $B_1 = \{\{1m\}, \quad \{1m, \quad 1m, \quad 1n\}\}$, $B_2 = \{\{1m\}, \quad \{2m, \quad 1n\}\}$, and $B_3 = \{\{1m\}, \quad \{2m1n\}\}$. If the mechanism determines the matching pattern to be 2, then supply bundle $2m1n$ can be only matched with one demand bundle $2m$ and one demand for $1n$.

The pre-determined pricing rule is modified as follows:

Step 1'': Set a single matching pattern e.

Step 2'': Solve the integer program $\boldsymbol{IP5}(I, J, e)$ which maximizes the social welfare $\widehat{V}_3(I, J)$.

[7] Typically, a shipper's demand on the online freight platform is on one lane, which is a special case of our case with multi-unit indivisible demand on multiple lanes. To increase the applicability of the mechanism, we consider this general case directly.

$$\boldsymbol{IP5}(I,J,e):\max_{|\{\alpha_i(e),\beta_j(e)\}|}\widehat{V}_3(I,J,e)=\sum_{i\in I}\alpha_i(e)v_i(o_i)-\sum_{j\in J}\beta_j(e)c_j \tag{30}$$

$$s.t.\sum_{i\in\{i|o_i=o,i\in I\}}\alpha_i(e)=\sum_{j\in J}\beta_j(e)q^o\left(b_j,e\right),\forall o\in O \tag{31}$$

$$\alpha_i(e)\in\{0,1\},\forall i\in I \tag{32}$$

$$\beta_j(e)\in\{0,1\},\forall j\in J \tag{33}$$

In objective function (30), $\alpha_i(e)(\beta_j(e))$ denotes whether shipper i (carrier j) trades under the matching pattern e. Constraint (31) guarantees supply and demand balance, in which $q^o\left(b_j,e\right)$ is the number of demand bundle o matched by supply bundle b_j under the matching pattern e. Let $\{\left(\alpha_i^*(e)\right)_{i\in I},\ \left(\beta_j^*(e)\right)_{j\in J}\}$ be the optimal solution to $\boldsymbol{IP5}(I,J,e)$.

Steps 3''-4'': Let $\widetilde{B}_e=\left\{b\middle|\sum_{j\in\{j|b_j=b,j\in J\}}\beta_j^*(e)\geq 1,b\in B\right\}$ and $B_e^*=\left\{b\middle|\sum_{j\in\{j|b_j=b,j\in J\}}\beta_j^*(e)\geq 2,b\in B\right\}$. Then these steps are the same as Steps 2–3 in the OBDA mechanism.

Step 5'': Let $O_e^*=\left\{o\middle|\sum_{i\in\{i|o_i=o,i\in I\}}\alpha_i^*(e)>\sum_{b\in\widetilde{B}_e}q^o(b,e),o\in O\right\}$. The pre-buying price for shipper i who bids for bundle $o_i\in O/O_e^*$ is set as $f_I^{o_i}(I,J)=0$.

Step 6'': Sort the bid prices of the shippers with $\alpha_i^*(e)=1$ for each demand bundle $o\in O$ in a strictly decreasing order. Let i_o^* be the shipper with the $\sum_{b\in\widetilde{B}_e}q^o(b,e)-th$ lowest bid price for demand bundle o. For each shipper i with $o_i=o$, its pre-buying price given I and J is $f_I^{o_i}(I,J)=\widehat{v}_{i_o^*}(o)$.

Note that we can consider the shippers' demand bundles as different transport lanes, i.e., a shipper's demand bundle o can be considered as a unit of transport service on lane "o". Then the modified pre-determined pricing rule is similar to the original one but follows the given matching pattern.

Remark 3 The OBDA-SMP mechanism is robust enough to make it also applicable in the case with multi-unit divisible demand and the case where a shipper can procure a bundle of services from multiple carriers. In the former case, we can treat shipper i's demand as an indivisible demand bundle o_i and the total bid price $\widehat{v}_iq_i$ as the bid price $\widehat{v}_i(o_i)$ for this bundle. Then the allocation scheme generated by the OBDA-SMP mechanism is still viable for these two cases but may lose some efficiency because a shipper can only trade with one carrier. In addition, the OBDA-SMP mechanism is equivalent to the OBDA mechanism in the case with single-unit demand because each supply bundle has only one matching pattern for the demand bundles.

Now we use a simple example to illustrate how the OBDA-SMP mechanism works.

Example 4 Table 5 shows the bids and asks from the agents in each period. Note that each shipper procures either all its demands or nothing. We assume $K=2$. In this example, $|E|=4$ and there are four available matching patterns, i.e., $B_1=\{\{1m,1n,1n\},\{1m,1m,1n\}\}$, $B_2=\{\{1m,1n,1n\},\{2m,1n\}\}$, $B_3=\{\{1m,2n\},\{1m,1m,1n\}\}$, and $B_4=\{\{1m,2n\},\{2m,1n\}\}$.

By solving $\boldsymbol{IP4}\left(I^{\leq 2},J^{\leq 2}\right)$, we can get the optimal allocation, i.e., shippers 1 and 7 procure transport services from carrier 1, shippers 3 and 5 procure transport services from carrier 3, and shippers 4 and 7 procure transport services from carrier 4 in period 0; shippers 10 and 11 procure transport services from carrier 5 in period 1; shippers 8 and 9 procure transport services from carrier 6, and shippers 12 and 13 procure transport services from carrier 7 in period 2. Then the ex-post maximum social welfare is 24.5 $(\underbrace{(8+12+10+7+6.5+14-16-15-16)}_{period\ \ 0}+\underbrace{(11+7-14)}_{period\ \ 1}+\underbrace{(13+9+6+14-15-17)}_{period\ \ 2}=24.5)$.

Then we employ the OBDA-SMP mechanism to clear the market. We first determine a matching pattern $e=4$, i.e., one supply bundle $1m2n$ can only match one demand bundle $1m$ and one demand bundle $2n$, and one supply bundle $2m1n$ can only match one demand bundle $2m$ and one demand bundle $1n$.

In period 0, by Eqs.(13)-(14), the admission prices for all shippers are 0 and for all carriers are ∞, respectively. All the agents are active. Next we use the pre-determined pricing rule to determine each agent's pre-buying (pre-selling) price in period 0. Specifically, by solving $\boldsymbol{IP5}(\vec{I}^0,\vec{J}^0,4)$, all shippers and carriers trade in the optimal allocation. According to **Steps 4''** and **6''** of the pre-determined pricing rule, for supply bundle $1m2n$, j_{1m2n}^* is carrier 2; for supply bundle $2m1n$, j_{2m1n}^* is carrier 4; for demand bundle $1m$, i_{1m}^* is shipper 2; for demand bundle $2m$, i_{2m}^* is shipper 4; for demand bundle $1n$, i_{1n}^* is shipper 6; and for demand bundle $2n$, i_{2n}^* is shipper 8. Then we have that the pre-buying price $f_I^{1m}(\vec{I}^0,\vec{J}^0)$ for each shipper i with $o_i=1m$ is 6, $f_I^{2m}(\vec{I}^0,\vec{J}^0)$ for each shipper i with $o_i=2m$ is 10, $f_I^{1n}(\vec{I}^0,\vec{J}^0)$ for each shipper i with $o_i=1n$ is 6.5, $f_I^{2n}(\vec{I}^0,\vec{J}^0)$ for each shipper i with $o_i=2n$ is 13. The pre-selling price $f_J^{1m2n}(\vec{I}^0,\vec{J}^0)$ for each carrier j with $b_j=1m2n$ is 18, and $f_J^{2m1n}(\vec{I}^0,\vec{J}^0)$ for each carrier j with $b_j=2m1n$ is 16. According to the allocation rule, the bid prices of shippers 1, 3, 5, and 7 are higher than their pre-buying prices and the ask prices of carriers 1 and 3 are lower than their pre-selling prices, so they can trade in this period. Other agents (shippers 2, 4, 6, 8, and carriers 2, 4) are rejected. According to the payment rule, the payment of shipper 1 is $\max\left\{p_i=0,f_I^{1m}(\vec{I}^0,\vec{J}^0)=6\right\}=6$. Similarly, shippers 3, 5, and 7 pay 6, 10, 6.5, and 13 for their demands respectively, and carriers 1 and 3 get 18 and 16 for their supplies, respectively. Since there are no shippers with a pre-buying price of 0 and no carriers with a pre-selling price of ∞, no bids and asks survive to the next period.

In period 1, we first calculate the admission prices for the newly arrived agents. For shipper 9, $\widehat{d}_9-\widehat{a}_9=2-1<K=2$ and then

Table 5
Bids and asks from the carriers in each period.

Period	Shipper:$\{\widehat{a}_i, \widehat{d}_i, \widehat{v}_i(o_i)\}$				Carrier:$\widehat{\theta}_j = \{\widehat{a}_j, \widehat{d}_j, \widehat{c}_j\}$	
	$1m$	$2m$	$1n$	$2n$	$1m2n$	$2m1n$
0	1:{0,1,8}	3:{0,0,12}	5:{0,0,7}	7:{0,1,14}	1:{0,1,16}	3:{0,1,15}
	2:{0,2,6}	4:{0,0,10}	6:{0,1,6.5}	8:{0,2,13}	2:{0,2,18}	4:{0,0,16}
1	9:{1,2,9}	10:{1,3,11}	11:{1,3,7}			5:{1,1,14}
2	12:{2,4,6}			13:{2,3,14}	6:{2,3,15}	
				14:{2,4,11}	7:{2,4,17}	

$p_9 = \max_{\tau\in[\max(0,\hat{d}_i-K),\hat{a}_i-1]}\left\{f_I^{1m}(\bar{I}^\tau \cup \{shipper\ 9\}, \bar{J}^\tau)\right\} = f_I^{1m}(\bar{I}^0 \cup \{shipper\ 9\}, \bar{J}^0)$ according to Eq.(13). By solving $\boldsymbol{IP5}(\bar{I}^0 \cup \{shipper\ 9\}, \bar{J}^0, 4)$, all agents except shipper 2 trade in the optimal allocation. According to **Step 6′′** of the pre-determined pricing rule, for demand bundle $1m$, i^*_{1m} is shipper 1 and then shipper 9′s admission price is $p_9 = f_I^{1m}(\bar{I}^0 \cup \{shipper\ 9\}, \bar{J}^0) = \widehat{v}_1 = 8$. Similarly, from Eqs.(13)-(14), the admission prices for shippers 10–11 are 0 and for carrier 5 is 15, respectively. Then, all the agents are active. From the pre-determined pricing rule, all the shippers' pre-buying prices are 0 and all the carriers' pre-selling prices are ∞ in this period. So they do not trade in this period but will remain active in the next period except for carrier 5. Carrier 5 will leave the market after period 1.

In period 2, the agent set contains the newly arrived agents (shippers 12–14 and carriers 6–7) and the previous surviving agents (shippers 9–11). The admission prices are 0, 0, and 0 for shippers 12–14, and ∞ and ∞ for carriers 6–7, respectively. All the agents are active in period 2. From the pre-determined pricing, we have $f_I^{1m}(\bar{I}^2, \bar{J}^2) = 6$, $f_I^{2m}(\bar{I}^2, \bar{J}^2) = 0$, $f_I^{1n}(\bar{I}^2, \bar{J}^2) = 0$, $f_I^{2n}(\bar{I}^2, \bar{J}^2) = 11$, and $f_J^{1m2n}(\bar{I}^2, \bar{J}^2) = 17$. Shippers 9 and 13 pay 8 and 11 for their demand, respectively. Carrier 6 gets 17 for its supply.

By the end of period 2, seven among the 16 shippers and three among the seven carriers have successfully traded. The shippers' total utility is 9.5 ($\underbrace{(8+12+7+14-6-10-6.5-13)}_{period\ 0} + \underbrace{(9+14-8-11)}_{period\ 2} = 9.5$) and the carriers' total utility is 5 ($\underbrace{(18+16-16-15)}_{period\ 0} + \underbrace{(17-15)}_{period\ 2} = 5$). The platform achieves an exact match between the supply and demand in each period and obtains a profit of 3.5 ($\underbrace{(6+10+6.5+13-18-16)}_{period\ 0} + \underbrace{(8+11-17)}_{period\ 2} = 3.5$). The OBDA-SMP mechanism yields the social welfare of 18 ($\underbrace{(8+12+7+14-18-16)}_{period\ 0} + \underbrace{(9+14-17)}_{period\ 2} = 18$) and realizes the efficiency of 73.47 % (18/24.5 = 73.47%). The example ends here.

The OBDA-SMP mechanism has the following properties.

Theorem 7 T*he OBDA-SMP mechanism is strongly feasible, ex-post weakly budget-balanced, ex-post individually rational, and incentive-compatible*.

Proof. Please see Appendix A.7.

Theorem 7 shows that the OBDA-SMP mechanism is practical for the general dynamic transport service transactions, which can incentivize the platform to hold auctions and can also attract and retain shippers and carriers to participate in the online platform transactions.

Unfortunately, we cannot further evaluate the efficiency loss bound of the OBDA-SMP mechanism, like OBDA and OBDA-padding mechanisms, because the efficiency loss is associated with the given matching pattern. In addition, because the matching pattern is pre-determined, it is not easy for the OBDA-SMP mechanism to achieve the same efficiency as that of the other two mechanisms.

Under the OBDA-SMP mechanism, the platform needs to pre-determine the matching pattern in period 0. An inappropriate matching pattern may lead to a large efficiency loss. Here we propose a feasible approach for the platform to select an appropriate matching pattern e. According to the transaction history, the platform can accumulate some bidding data about the agents. Then the platform can set e to be the optimal matching pattern in the optimal solution to the following nonlinear integer program, which maximizes the social welfare under the single matching pattern.

$$\boldsymbol{INLP1}(I,J): \max_{[\gamma(e),\alpha_i(e),\beta_i(e)]} V_2(I,J) = \sum_{e\in E}\gamma(e)\left[\sum_{i\in I}\alpha_i(e)v_i(o_i) - \sum_{j\in J}\beta_j(e)c_j\right] \tag{34}$$

$$s.t. \sum_{e\in E} r(e) \le 1 \tag{35}$$

$$\sum_{i\in\{i|o_i=o, i\in I\}}\alpha_i(e) = \sum_{j\in J}\beta_j(e)q^o(b_j,e), \forall o\in O, \forall e\in E \tag{36}$$

$$\gamma(e)\in\{0,1\}, \forall e\in E \tag{37}$$

$$\alpha_i(e)\in\{0,1\}, \forall i\in I, \forall e\in E \tag{38}$$

$$\beta_j(e) \in \{0,1\}, \forall j \in J, \forall e \in E \tag{39}$$

In objective function (34), $\gamma(e)$ denotes whether the matching pattern e is selected. Constraint (35) means that at most one matching pattern can be selected. Constraints (37) stipulate that $\gamma(e)$ is a binary variable.

Note that it is difficult to find the optimal solution exactly to the nonlinear integer optimization problem ***INLP1***(I,J). To circumvent this difficulty, we transform ***INLP1***(I,J) into the following integer program ***IP6***(I,J) by some linearization techniques, and **Proposition 1** shows that ***IP6***(I,J) is equivalent to ***INLP1***(I,J). Then we can solve ***IP6***(I,J) exactly by existing algorithms.

$$\textbf{\textit{IP6}}(I,J): \max_{[\gamma(e),\alpha_i(e),\beta_i(e),\eta_i(e),\kappa_j(e)]} \widetilde{V}_2(I,J) = \sum_{e\in E}\left[\sum_{i\in I}\eta_i(e)v_i(o_i) - \sum_{j\in J}\kappa_j(e)c_j\right] \tag{40}$$

$$s.t. \sum_{e\in E} r(e) \le 1 \tag{41}$$

$$\sum_{i\in\{i|o_i=o, i\in I\}} \alpha_i(e) = \sum_{j\in J}\beta_j(e)q^o(b_j,e), \forall o \in O, \forall e \in E \tag{42}$$

$$\alpha_i(e) \le \gamma(e), \forall i \in I, \forall e \in E \tag{43}$$

$$\eta_i(e) \le \alpha_i(e), \forall i \in I, \forall e \in E \tag{44}$$

$$\gamma(e) + \alpha_i(e) - \eta_i(e) \le 1, \forall i \in I, \forall e \in E \tag{45}$$

$$\beta_j(e) \le \gamma(e), \forall j \in J, \forall e \in E \tag{46}$$

$$\kappa_j(e) \le \beta_j(e), \forall j \in J, \forall e \in E \tag{47}$$

$$\gamma(e) + \beta_j(e) - \kappa_j(e) \le 1, \forall j \in J, \forall e \in E \tag{48}$$

$$\eta_i(e) \in \{0,1\}, \forall i \in I, \forall e \in E \tag{49}$$

$$\kappa_j(e) \in \{0,1\}, \forall j \in J, \forall e \in E \tag{50}$$

$$\gamma(e) \in \{0,1\}, \forall e \in E \tag{51}$$

$$\alpha_i(e) \in \{0,1\}, \forall i \in I, \forall e \in E \tag{52}$$

$$\beta_j(e) \in \{0,1\}, \forall j \in J, \forall e \in E \tag{53}$$

In objective function (40), $\eta_i(e)$ and $\kappa_j(e)$ are auxiliary decision variables.

Proposition 1 *The optimal solution to* ***IP6***(I,J) *is an optimal solution to* ***INLP1***(I,J).

Proof. Please see Appendix A.8.

6. Numerical studies

In this section we conduct numerical studies to assess the efficiency of the three mechanisms and examine the impacts of market size and agents' patience on their performance. We include the code and data of the simulation in the GitHub repository found at https://github.com/Guojianto/-Truthful-Online-Bundle-Double-Auction-Mechanisms.git.

6.1. Experimental parameter settings

Consider a transport network of the 15 provincial capitals in North, Central, and East China.[8] Carriers provide transport services over this transport network with the homogenous 9.6-meter vehicles with a 13.5-tonne load. The agents dynamically arrive at and depart from the platform over time periods to trade transport services on 20 lanes that are randomly generated among the 15 cities. Each shipper wants to procure at most two truckloads on one lane. Each carrier can supply a bundle of transport services on these lanes. The number of lanes in each bundle is not larger than two, and the number of truckloads on each lane is independently drawn from the discrete uniform distribution in the interval [1, 2]. The carriers' transport service costs (\$/ton·km) on the lanes in their supply bundles follow the continuous uniform distribution $U_D[0.03, 0.05]$(Guo et al., 2022). To cover the carriers' costs, the shippers' valuations (\$/ton·km) for transport service on the lanes follow the continuous uniform distribution $U_D[0.036, 0.06]$, which is 20 % higher than the

[8] Note that our numerical studies are not limited to the dataset generated from this China transport network. In Appendix A.9 we conduct numerical studies on a dataset generated from a transport network of five hub cities in the U.S. and draw similar findings as in this section.

carriers' costs.

We consider there are eight periods in the experiments. The number of bundle types, the number of agents, and agents' patience in each period are set differently in different studies. For each set of parameter settings, we simulate the OBDA, OBDA-padding, and OBDA-SMP mechanisms to allocate the transactions and set the transaction prices, respectively. The other settings for each mechanism are as follows: (i) under the OBDA mechanism, each shipper only procures one truckload of transport services on one lane (corresponding to the case with single-unit demand); (ii) under the OBDA-SMP mechanism, each shipper buys either all its demand or nothing (corresponding to the case with multi-unit indivisible demand on multiple lanes); and (iii) note that a shipper will buy less under the OBDA mechanism, to balance demand and supply, the number of shippers is three times the number of carriers under the OBDA mechanism and the number of shippers is twice the number of carriers under OBDA-padding and OBDA-SMP mechanisms.[9]

All mechanisms are programmed in Matlab with the runtime environment as follows: Windows 11, Intel Core i5-1240P CPU @ 1.70 GHz, and 16 GB RAM. The integer optimization problems in our auction mechanisms are solved by calling Gurobi via Matlab. We repeat the randomized simulation test 20 times for each set of parameter settings and report the average values.[10]

6.2. The impacts of the market size

In this subsection we study the impacts of market size on the performance of our mechanisms. The numbers of bundle types and the carriers for each bundle type are used to measure the market size. We set the number of bundle types to increase from 10 to 16 in steps of three and set the number of carriers for each bundle type to increase from 20 to 40 in steps of ten. We set K to one, which means that the agents can stay in the market for up to two periods. Table 6 reports the results, in which "Efficiency" stands for the efficiency achieved by each mechanism, "Agents" for the proportion of social welfare for all agents, and "Platform" for the platform's profit.

From Table 6, the efficiencies of our mechanisms increase with the market size, in terms of the numbers of both supply bundle types and carriers for each bundle. When the market is large, the mechanisms' efficiencies can be more than 90 %. In terms of the carrier number for each bundle, it is larger, more agents can eventually trade. From **Theorems 3 and 6**, more trading agents lead to smaller efficiency loss, so the OBDA and OBDA-padding mechanisms can be more efficient in a market with a larger number of agents. The OBDA-SMP mechanism has a similar trend. In terms of the number of supply bundle types, more bundle types may result in more efficiency loss but increase the quantity of demand for transportation services on each lane because the lane number is fixed (20 lanes). Then for each lane, there will be more trading shippers. This can increase the efficiency and thus can offset the efficiency loss due to the increased bundle types. So the efficiencies of our mechanisms also increase with the number of supply bundle types.

Among the three mechanisms, the OBDA mechanism is the most efficient, followed by the OBDA-padding mechanism and finally the OBDA-SMP mechanism. This is consistent with what we mentioned earlier. The OBDA-padding mechanism removes more profitable transactions due to padding and allocating transactions according to bidding times, so it loses more efficiency compared to the OBDA mechanism. The OBDA-SMP mechanism effectuates the lowest efficiency because it allocates transactions under a fixed matching pattern.

In terms of social welfare division, our mechanisms tend to allocate more social welfare to the agents. As the market size increases, the agents' share of social welfare also increases. When there are 16 bundle types and 40 carriers for each bundle type, the agents can get more than 78 % of the social welfare. This is because a larger market leads the mechanisms to set the shippers' buying price and the carriers' selling price closer to the market equilibrium price, thus narrowing the buy-sell spread. Then the share of the social welfare allocated to the platform is compressed, while that to the agents is increased. This can make the agents aware that the platform does not grab most of the welfare at the broker's advantage. As a result, agents are more willing to participate in online platform transactions. In addition, although the proportion of social welfare occupied by the platform decreases, the platform's total profit increases. This can give the platform an incentive to attract more agents to participate the platform.

Overall, our mechanisms give the platform an incentive to expand the transaction market because it could kill three birds with one stone by (i) improving efficiency, (ii) earning more profit, and (iii) attracting and retaining more agents.

6.3. The impacts of agents' patience

In this subsection we study the impacts of agents' patience on the performance of our mechanisms. We set the agents' patience to increase from 0 to 3 in the step of 1. When the patience is zero, the auction is reduced to a repeated static auction. We consider that there are 13 bundle types and 30 carriers that want to provide transport services for each bundle type in each period. Table 7 reports the results.

From Table 7, we find that the efficiencies of our mechanisms decrease with the agents' patience bound. This is inconsistent with our intuition. Intuitively, the longer the agents stay, the more efficient the market should be, because it gives the market flexibility to

[9] Under the OBDA mechanism, each shipper has one unit of demand on one lane. For a given bundle, it includes up to two lanes and the supply on each lane is independently drawn from $U[1,2]$. Thus the expected transport supply in a given bundle is 2.25 units. We consider a market where demand slightly outstrips supply and thus set the number of shippers to be three times the number of carriers under the OBDA mechanism. Under the OBDA-padding and OBDA-SMP mechanisms, a shipper's demand is independently drawn from $U[1,2]$ with an expectation of 1.5. To compare with the OBDA mechanism in the same benchmark, we consider the same total demand as in the case of the OBDA mechanism and thus set the number of shippers to be twice the number of carriers under the latter two mechanisms.

[10] We provide a robustness check on the number of experiments in Appendix A.10 and find that our results are robust.

Table 6
The performance of the mechanisms at different market sizes.

	OBDA mechanism			OBDA-padding mechanism			OBDA-SMP mechanism		
	Efficiency	Agents	Platform	Efficiency	Agents	Platform	Efficiency	Agents	Platform
10 bundle types									
20	88.95 %	66.91 %	$92,472	84.12 %	66.50 %	$78,327	70.76 %	59.99 %	$60,961
30	93.92 %	75.34 %	$110,035	90.81 %	75.27 %	$104,198	78.62 %	67.79 %	$92,078
40	95.53 %	78.93 %	$127,111	92.86 %	77.26 %	$124,503	83.73 %	70.94 %	$113,819
13 bundle types									
20	91.73 %	71.79 %	$126,109	88.34 %	72.96 %	$103,741	76.57 %	65.53 %	$88,586
30	94.92 %	77.41 %	$146,637	93.01 %	77.57 %	$135,240	84.17 %	71.66 %	$125,774
40	96.74 %	81.91 %	$161,230	95.19 %	81.91 %	$140,909	88.06 %	75.07 %	$139,631
16 bundle types									
20	92.30 %	71.72 %	$158,351	90.11 %	72.66 %	$126,329	83.24 %	66.76 %	$110,017
30	95.97 %	79.59 %	$190,237	93.71 %	78.86 %	$175,812	87.50 %	73.95 %	$163,953
40	97.14 %	82.64 %	$213,459	95.85 %	82.92 %	$198,507	91.05 %	78.43 %	$192,569

Table 7
The performance of the mechanisms with different agents' patience.

K	OBDA mechanism			OBDA-padding mechanism			OBDA-SMP mechanism		
	Efficiency	Agents	Platform	Efficiency	Agents	Platform	Efficiency	Agents	Platform
0	99.37 %	89.30 %	$75,269	96.90 %	89.20 %	$70,751	92.77 %	84.21 %	$76,798
1	94.92 %	77.41 %	$146,637	93.01 %	77.57 %	$135,240	84.17 %	71.66 %	$125,774
2	88.38 %	64.88 %	$213,298	85.85 %	64.10 %	$206,086	72.29 %	57.03 %	$162,856
3	80.84 %	57.11 %	$246,401	78.81 %	57.40 %	$224,624	61.19 %	51.99 %	$148,625

time trades by receiving the demand and supply in one period but releasing them in later periods. But this is not the case with our mechanisms. The main reason is that to incentivize agents to submit their arrival and departure times truthfully, our mechanisms set the admission prices and transaction prices considering several periods before the arrivals of the agents as shown in Eqs.(13) and (14). Then our mechanisms tend to set higher admission and buying prices for the shippers but lower admission and selling prices for the carriers as K increases. This will lead to a larger gap between the shippers' payments and the carriers' payoffs, which will reduce the transaction quantities. As a result, the efficiency of the mechanisms is reduced. However, the efficiency of the mechanisms is very high when the agents are impatient (all are larger than 90 %).

In addition, we find that as the agents' patience increases, the platform's profit increases under the OBDA and OBDA-padding mechanisms, and it first increases and then decreases under the OBDA-SMP mechanism. It comes down to the fact that the large transaction price gap is a double-edged sword for the platform. A large transaction price gap means a high marginal profit for the platform but a low trading quantity. For the OBDA-SMP mechanisms, when $K \leq 2$, the positive effect from increasing K outweighs the negative one. Therefore, as K increases, the platform's profit increases. When $K > 2$, the negative effect dominates and the platform's profit decreases. For the OBDA and OBDA-padding mechanisms, the negative effect is dominated by the positive one, so the platform's profit increases with the agents' patience. However, the incremental profit of the platform decreases. Combined with the case of the OBDA-SMP mechanism, we can predict that the platform's profit will also fall under these two mechanisms when K is large enough. This finding implies that the platform needs to balance efficiency and profit by setting agents' maximum dwell time in the transaction.

Table 8
The average runtime (second) per period of the three mechanisms at different market sizes.

	OBDA mechanism	OBDA-padding mechanism	OBDA-SMP mechanism
10 bundle types			
20	2.68	2.17	2.16
30	4.67	3.79	3.64
40	6.41	6.03	5.69
13 bundle types			
20	4.06	3.19	3.08
30	6.45	5.59	5.08
40	10.39	8.75	7.45
16 bundle types			
20	5.12	4.05	4.15
30	9.00	7.47	7.26
40	14.51	11.56	9.72

6.4. Computational efficiency analysis of the mechanisms

In this subsection, we analyze the computational efficiency of our mechanisms. We also record the runtimes of each mechanism for different test cases. Tables 8 and 9 record the average runtimes (second) per period of the three mechanisms for different parameter settings in Subsections 6.2 and 6.3.

From Tables 8 and 9, the OBDA-SMP mechanism takes the shortest running time, followed by the OBDA-padding mechanism, and finally the OBDA mechanism. This is because the OBDA-SMP mechanism makes pricing and allocation decisions by solving $\boldsymbol{IP5}(I,J,e)$. $\boldsymbol{IP5}(I,J,e)$ is constrained by a single matching pattern, which reduces the range of feasible domains. Thus the OBDA mechanism has the highest computational efficiency. As for the other two mechanisms, the OBDA and OBDA-padding mechanisms solve 0–1 integer program $\boldsymbol{IP2}(I,J)$ and integer program $\boldsymbol{IP3}(I,J,\boldsymbol{S})$ to make the pricing and allocation decisions, respectively. Given the same quantities of supply and demand, the feasible region of $\boldsymbol{IP2}(I,J)$ is larger than that of $\boldsymbol{IP3}(I,J,\boldsymbol{S})$. For example, consider the case with three shippers with a single unit of demand in $\boldsymbol{IP2}(I,J)$ and a shipper with three units of demand in $\boldsymbol{IP3}(I,J,\boldsymbol{S})$, respectively. The are more feasible solutions ($\{0,1\}\times\{0,1\}\times\{0,1\}$) in $\boldsymbol{IP2}(I,J)$ than that ($\{0,1,2,3\}$) in $\boldsymbol{IP3}(I,J,\boldsymbol{S})$. Therefore, the computational efficiency of the OBDA-padding mechanism is higher than that of the OBDA mechanism.

From Table 8, we can further find that the runtimes of the three mechanisms increase with the market size. Especially as the number of the carriers for each bundle type increases, the runtimes increase quickly. This is because some decisions such as pricing and allocation under our mechanisms require solving some integer programming problems, for example, $\boldsymbol{IP2}(I,J)$. An increase in the market size will obviously increase the numbers of decision variables and constraints in the integer programming problems. Then the computational time required will also increase, hence increasing the runtime. For the number of bundle types, the market size increases linearly with it, and then the runtime of the mechanisms increases more slowly. Whereas for the number of carriers for each bundle type, the market size increases exponentially with it, and thus the runtime required by the mechanisms increases significantly.

From Table 9, we can find that the runtimes of the three mechanisms increase nearly linearly with the agents' patience when $K>0$. This is because that in our mechanisms, the determination of each agent's admission price considers pre-determined pricing schemes for $(\widehat{a}_i-\widehat{d}_i+K)$ periods before the agent arrives from Eqs.(13) and (14). The determination of the pre-determined price for one agent per period requires solving an integer programming problem. As K increases, the number of integer programming problems to be solved increases linearly. Therefore, the runtime of our mechanisms increases nearly linearly with the agents' patience.

Nonetheless, the computation time per period required by our mechanisms is not large. For example, even in a large transport market (16 bundle types and 40 carriers for each bundle type), the average runtime required by the OBDA mechanism is about 14.51 s. In FTL transport service transactions, the length of a clearing period is usually not very short, so our mechanism can fully satisfy the platform's requirement for clearing time in practice.

7. Conclusion and future research

In this paper, we study the mechanism design problem for the transport market where shippers and carriers dynamically arrive and depart over time to trade their transport services over a transport network. The agents are self-interested and their valuations, costs, and arrival and departure times are private information. An online freight platform, as a third-party auctioneer, holds the online bundle double auctions to allocate the transactions and sets the transaction prices in each period.

We first consider the case where each shipper wants a single unit of transport services on one lane, while the supply of each carrier is a bundle of transport services. We propose the OBDA mechanism for this case and show that the mechanism is feasible, incentive-compatible, individually rational, and budget-balanced. We further consider two more general cases and propose two variants of the OBDA mechanism for them by modifying some rules of the OBDA mechanism: namely, the OBDA-padding mechanism for the case with multi-unit demand on one lane and the OBDA-SMP mechanism for the case with multi-unit indivisible demand on multiple lanes. We show that both modified mechanisms are feasible, incentive-compatible, individually rational, and budget-balanced. In addition, the OBDA-SMP mechanism has more applications (e.g., the case with multi-unit demand on one lane) but less efficiency than the OBDA and OBDA-padding mechanisms. We conduct numerical studies to investigate the impacts of market size and agents' patience on the performance of the mechanisms. We obtain the following findings: (i) The large market size can improve the mechanisms' efficiency, increase the platform's profit, and attract and retain more agents; (ii) As the agents' patience increases, the efficiency of our mechanisms decreases, while the platform's profit first increases and then decreases; (iii) the mechanisms can get high efficiency when the agents are impatient; and (iv) the computational efficiency of our mechanisms can satisfy the platform's requirement for clearing time in practice.

Based on this study, there are some topics need to be studied further.

In our mechanisms, we impose a bidding restriction on the carriers, i.e., a carrier can only ask for one bundle of transport services.

Table 9
The average runtime (second) per period of three mechanisms with different agents' patience.

K	OBDA mechanism	OBDA-padding mechanism	OBDA-SMP mechanism
0	0.02	0.02	0.02
1	6.45	5.59	5.08
2	13.12	10.28	8.87
3	19.34	14.75	13.37

This would decrease the flexibility of the carriers' biddings, thereby reducing the chance of successful transactions. However, if we allow a carrier to offer multiple bundles of transport services but ultimately win only one (also called XOR bidding), our mechanisms hardly ensure incentive compatibility. This is because it is difficult to prevent the carriers from manipulating their bid prices for different bundles to influence the ultimately sold bundle to get a higher utility. Even in the static bilateral trading market, this is a well-known hard problem (Xu et al., 2016; Sun et al., 2019). Thus, it is an interesting and challenging problem to design a truthful online mechanism for this case.

Another interesting topic is to improve the efficiency of our online bundle double auction mechanisms. On the one hand, our mechanisms lose some efficiency to ensure the truthfulness with respect to the arrival and departure information. Therefore, the efficiency is not very high when the agents have a large patience bound. A challenging topic is to find a way to overcome the conflict between the mechanism efficiency and truthfulness and then design a more efficient online mechanism for our problem. On the other hand, the OBDA-SMP mechanism has a large efficiency loss due to a given single matching pattern. Therefore, another meaningful topic is to relax this single matching pattern restriction and to design a truthful and more efficient mechanism for the case with multi-unit indivisible demand on multiple lanes.

CRediT authorship contribution statement

Jiantao Guo: Writing – original draft, Methodology, Conceptualization. **Juliang Zhang:** Supervision. **T.C.E. Cheng:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Proof of Theorem 1

Strong feasibility. Let $\left\{\left(\alpha_i^{t^*}\right)_{i\in\vec{I}^t},\left(\beta_j^{t^*}\right)_{j\in\vec{J}^t}\right\}$ be the optimal solution to ***IP2***$(\vec{I}^t,\vec{J}^t)$. We have $\sum_{i\in\{i|l_i=l,i\in\vec{I}^t\}}\alpha_i^{t^*}=\sum_{j\in\vec{J}^t}\beta_j^{t^*}q^l(b_j)$ for each $l\in L$ from constraint (10). Let $B^{t^*}=\left\{b\middle|\sum_{j\in\{j|b_j=b,j\in J\}}\beta_j^{t^*}\geq 1,b\in B\right\}$. In period t, the shippers with $\widehat{v}_i>f_I^{l_i}(\vec{I}^t,\vec{J}^t)$ and the carriers with $\widehat{c}_j<f_J^{b_j}(\vec{I}^t,\vec{J}^t)$ will finally trade. Then, for each bundle $b\in B^{t^*}$, there are $\sum_{j\in\{j|b_j=b,j\in\vec{J}^t\}}\beta_j^{t^*}-1$ asks being traded and $\sum_{j\in\{j|b_j=b,j\in\vec{J}^t\}}y_j^t=\sum_{j\in\{j|b_j=b,j\in\vec{J}^t\}}\beta_j^{t^*}-1$. For each lane l, there are $\sum_{i\in\{i|l_i=l,i\in\vec{I}^t\}}\alpha_i^{t^*}-\sum_{b\in B^{t^*}}q^l(b)$ units being traded and $\sum_{i\in\{i|l_i=l,i\in\vec{I}^t\}}x_i^t=\sum_{i\in\{i|l_i=l,i\in\vec{I}^t\}}\alpha_i^{t^*}-\sum_{b\in B^{t^*}}q^l(b)$. Then, for each lane $l\in L$, we have

$$\sum_{i\in\{i|l_i=l,i\in\vec{I}^t\}}x_i^t=\sum_{i\in\{i|l_i=l,i\in\vec{I}^t\}}\alpha_i^{t^*}-\sum_{b\in B^{t^*}}q^l(b)=\sum_{j\in\vec{J}^t}\beta_j^{t^*}q^l(b_j)-\sum_{b\in B^{t^*}}q^l(b)=\sum_{b\in B^{t^*}}\left(\sum_{j\in\{j|b_j=b,j\in\vec{J}^t\}}\beta_j^{t^*}-1\right)q^l(b_j)$$
$$=\sum_{b\in B^{t^*}}\sum_{j\in\{j|b_j=b,j\in\vec{J}^t\}}y_j^tq^l(b_j)=\sum_{j\in\vec{J}^t}y_j^tq^l(b_j)$$

In each period t, the demand and supply are exactly balanced. Thus, the OBDA mechanism is *strongly* feasible.

Ex-post weak budget balance. From the pre-determined pricing rule, we have $\widehat{c}_{j_b^*}\leq\sum_{l\in L}q^l(b)\widehat{v}_{i_l^*}$ for each bundle $b\in B^{t^*}$. In period t, the shippers with $\widehat{v}_i>\widehat{v}_{i_l^*}$ and the carriers with $\widehat{c}_j<\widehat{c}_{j_b^*}$ will trade. From the feasibility of the mechanism, the platform always has a non-negative profit in each period t. Thus, the OBDA mechanism is *ex-post weakly* budget-balanced.

Ex-post individual rationality. For trading shipper i, it must be active. So we have $\widehat{v}_i>p_i$. When it trades in period t, $\widehat{v}_i>f_I^{l_i}(\vec{I}^t,\vec{J}^t)$. The shipper's payment is $\pi_i^t=\max\left\{p_i,f_I^{l_i}(\vec{I}^t,\vec{J}^t)\right\}$. Therefore $\widehat{v}_i>\pi_i^t$ and it gets positive utility from truthful bidding.

Similarly, for trading carrier j, we have $\widehat{c}_j>p_j$ and $\widehat{c}_j<f_J^{b_j}(\vec{I}^t,\vec{J}^t)$ and it gets $\omega_i^t=\min\left\{p_j,f_J^{b_j}(\vec{I}^t,\vec{J}^t)\right\}$. Therefore $\widehat{c}_j<\omega_j^t$ and it gets a

positive utility from truthful bidding. Thus, the OBDA mechanism is *ex-post* individually rational. ***Q.E.D.***

Proof of Theorem 2

We first show that the agents will bid (ask) their true valuations (costs) under the OBDA mechanism.

Lemma EC.1. *The agents will bid their valuations and costs truthfully under the OBDA mechanism.*

Proof. If shipper i trades in period t, its payment π_i^t is $\max\left\{p_i, f_I^{l_i}(\vec{I}^t, \vec{J}^t)\right\}$. If $d_i - a_i = K$, $p_i = 0$, which is independent of $\widehat{v}_i$. If $d_i - a_i < K$, then $p_i = \max_{\tau\in[\max(0,\hat{d}_i-K),\hat{a}_i-1]} f_I^{l_i}(\vec{I}^\tau \cup \{i\}, \vec{J}^\tau)$. In each period $t \in [\max(0, \widehat{d}_i - K), \widehat{a}_i - 1]$, $f_I^{l_i}(\vec{I}^t \cup \{i\}, \vec{J}^t)$ is only dependent on the shipper set $\vec{I}^t \cup \{i\}$ and the carrier set $\vec{J}^t$ rather than the agent set in the previous or subsequent period. Then, by similar arguments as in (Xu et al., 2016), shipper i's dominant strategy is to bid $\widehat{v}_i = v_i$. Thus, the shippers will bid their valuations truthfully under the OBDA mechanism.

By similar arguments as above, we have that the carriers will ask their costs truthfully. ***Q.E.D.***

Now we shall show that the agents will join the auction as soon as they arrive at the market, i.e., $\widehat{a}_i = a_i$ for all i and $\widehat{a}_j = a_j$ for all j.

Lemma EC.2. *The agents will not delay joining the auction under the OBDA mechanism.*

Proof. If $d_i - a_i = K$, shipper i will not delay joining the auction because p_i is zero, which is the lowest price it will pay.

Then we consider the case that $d_i - a_i < K$. If shipper i can trade in period t while bidding truthfully, its payment is $\pi_i^t = \max\left\{p_i, f_I^{l_i}(\vec{I}^t, \vec{J}^t)\right\} = \max\left\{\max_{\tau\in[\max(0,d_i-K),a_i-1]} f_I^{l_i}(\vec{I}^\tau \cup \{i\}, \vec{J}^\tau), \max_{\tau\in[a_i,t]} f_I^{l_i}(\vec{I}^\tau, \vec{J}^\tau)\right\}$. If shipper i bids $a_i < \widehat{a}_i < t$, it can also trade in period t and its payment is $\max\left\{\max_{\tau\in[\max(0,\hat{d}_i-K),\hat{a}_i-1]} f_I^{l_i}(\vec{I}^\tau \cup \{i\}, \vec{J}^\tau), \max_{\tau\in[\hat{a}_i,t]} f_I^{l_i}(\vec{I}^\tau, \vec{J}^\tau)\right\}$, which is the same as it bids $\widehat{a}_i = a_i$. If shipper i bids $t' > \widehat{a}_i > t > a_i$, it may trade in period t' and pay $\max\left\{\max_{\tau\in[\max\{0,\hat{d}_i-K\},\hat{a}_i-1]} f_I^{l_i}(\vec{I}^\tau \cup \{i\}, \vec{J}^\tau), \max_{\tau\in[\hat{a}_i,t']} f_I^{l_i}(\vec{I}^\tau, \vec{J}^\tau)\right\}$, which is no less than $\max\left\{\max_{\tau\in[\max(0,d_i-K),a_i-1]} f_I^{l_i}(\vec{I}^\tau \cup \{i\}, \vec{J}^\tau), \max_{\tau\in[a_i,t]} f_I^{l_i}(\vec{I}^\tau, \vec{J}^\tau)\right\}$, its payment while bidding $\widehat{a}_i = a_i$. So shipper i cannot obtain higher utility by bidding late ($a_i < \widehat{a}_i$).

By similar arguments as above, we can show that a carrier cannot get higher utility by bidding late. Thus, the agent will not delay joining the auction under the OBDA mechanism. ***Q.E.D.***

Lemma EC.3 shows that the agents will not leave the market early, i.e., $\widehat{d}_i = d_i$ for each i and $\widehat{d}_j = d_j$ for each j.

Lemma EC.3. *The agents will not leave the market early under the OBDA mechanism.*

Proof. If $d_i - a_i = K$, shipper i will not leave the market early because p_i is zero, which is the lowest price it will pay.

Then we consider the case where $d_i - a_i < K$. If shipper i can trade in period t, its payment is $\pi_i^t = \max\left\{\max_{\tau\in[\max(0,\hat{d}_i-K),a_i-1]} f_I^{l_i}(\vec{I}^\tau \cup \{i\}, \vec{J}^\tau), \max_{\tau\in[a_i,t]} f_I^{l_i}(\vec{I}^\tau, \vec{J}^\tau)\right\}$ while bidding truthfully. If shipper i bids $\widehat{d}_i < t < d_i$, it will lose the trade opportunity in period t. If shipper i bids $t < \widehat{d}_i < d_i$, it can also trade in period t and its payment is $\max\left\{\max_{\tau\in[\max(0,\hat{d}_i-K),\hat{a}_i-1]} f_I^{l_i}(\vec{I}^\tau \cup \{i\}, \vec{J}^\tau), \max_{\tau\in[\hat{a}_i,t]} f_I^{l_i}(\vec{I}^\tau, \vec{J}^\tau)\right\}$, which is no less than $\max\left\{\max_{\tau\in[\max(0,d_i-K),\hat{a}_i-1]} f_I^{l_i}(\vec{I}^\tau \cup \{i\}, \vec{J}^\tau), \max_{\tau\in[\hat{a}_i,t]} f_I^{l_i}(\vec{I}^\tau, \vec{J}^\tau)\right\}$ its payment when $\widehat{d}_i = d_i$ because $\max_{\tau\in[\max(0,\hat{d}_i-K),\hat{a}_i-1]} f_I^{l_i}(\vec{I}^\tau \cup \{i\}, \vec{J}^\tau) \geq \max_{\tau\in[\max(0,d_i-K),\hat{a}_i-1]} f_I^{l_i}(\vec{I}^\tau \cup \{i\}, \vec{J}^\tau)$. So the shipper cannot attain higher utility by leaving early ($\widehat{d}_i < d_i$).

By similar arguments, we can also show that a carrier cannot get higher utility by leaving early. Then we prove the lemma. ***Q.E.D.***

From **Lemmas EC.1, EC.2**, and **EC.3**, the OBDA mechanism is incentive-compatible. ***Q.E.D.***

Proof of Theorem 3

Let Δ_τ be the social welfare loss of the mechanism in period τ. Define the efficiency loss *el* of the OBDA mechanism from period 0 to period t as $el(t) = E\left[\frac{\sum_{\tau=0}^{t}\Delta\tau}{V_1(I^{\leq t}, J^{\leq t})}\right]$. According to this definition, we have that $E[el(t)] = E\left[\frac{\sum_{\tau=0}^{t}\Delta\tau}{V_1(I^{\leq t}, J^{\leq t})}\right] \leq E\left[\sum_{\tau=0}^{t}\frac{\Delta\tau}{V_1(I^\tau, J^\tau)}\right]$, $I^\tau = \{i | a_i = \tau, i \in I^{\leq t}\}$ and $J^\tau = \{j | a_j = \tau, j \in J^{\leq t}\}$.

Let $I_\tau^l = \{i | l_i = l, i \in I^\tau\}$ and $J_\tau^b = \{j | b_j = b, j \in J^\tau\}$. To avoid confusion, we use $v_{i\tau}^l$ to indicate the valuation of shipper $i \in I_\tau^l$ for lane l and $c_{j\tau}^b$ to indicate the cost of carrier $j \in J_\tau^b$ for bundle b, respectively. We further need the following assumptions. Given lane l and period τ, $v_{i\tau}^l$ is drawn independently from a distribution G^l with continuous density g^l on $[\underline{v}^l, \overline{v}^l]$. Given bundle b and period τ, $c_{j\tau}^b$ is drawn independently from a distribution G^b with continuous density g^b on $[\underline{c}^b, \overline{c}^b]$. We also require that the density functions g^l and g^b have non-zero minimum values, i.e., $\min\{g^l(x) : \underline{v}^l \leq x \leq \overline{v}^l\} > 0$ and $\min\{g^b(x) : \underline{c}^b \leq x \leq \overline{c}^b\} > 0$.

Let $m_\tau^l = |I_\tau^l|$ and $n_\tau^b = |J_\tau^b|$. We assume that for each lane l, m_τ^l is drawn independently from a positive discrete distribution with an expectation of $\overline{m}^l$ and for each bundle b, n_τ^b is drawn independently from a positive discrete distribution with an expectation of $\overline{n}^b$. Sort the bid prices of the shippers in I_τ^l in descending order for given l. Without loss of generality, we assume that $v_{1\tau}^l \geq v_{2\tau}^l \geq \ldots \geq v_{m_\tau^l\tau}^l$. Sort the ask prices of the carriers in J_τ^b in ascending order for given b. Without loss of generality, we assume that $c_{1\tau}^b \leq c_{2\tau}^b \leq \ldots \leq c_{n_\tau^b\tau}^b$. Let k_τ^l and k_τ^b be the numbers of trading shippers for lane l and trading carriers for bundle b in the efficient allocation in period τ, respectively.

We assume that for each lane l, k_τ^l is drawn independently from a positive discrete distribution with an expectation of K^l and for each bundle b, k_τ^b is drawn independently from a positive discrete distribution with an expectation of K^b.

Then we can find the lower bound on the *ex-post* maximum social welfare as follows,

$$\sum_{\tau=0}^{t}\widehat{V}_1\left[\left(k_\tau^l\right)_{\tau\le t}^{l\in L},\left(k_\tau^b\right)_{\tau\le t}^{b\in B}\right]=\sum_{\tau=0}^{t}\left[\sum_{l\in L}\sum_{i=1}^{k_\tau^l}\left(v_{i\tau}^l-v_{k_\tau^l\tau}^l\right)+\sum_{b\in B}\sum_{j=1}^{k_\tau^b}\left(c_{k_\tau^b\tau}^b-c_{j\tau}^b\right)+\left(\sum_{l\in L}k_\tau^l v_{k_\tau^l\tau}^l-\sum_{b\in B}k_\tau^b c_{k_\tau^b\tau}^b\right)\right]$$

If $d-a=K$ for any agent, we have that $\overline{I}^\tau=I^\tau$ and $\overline{J}^\tau=J^\tau$. Then the welfare loss (for each b) of the OBDA mechanism is caused by the lost transactions between the K_b-th carrier and the shippers who traded with it. Then the total welfare loss is $\sum_{\tau=0}^{t}\Delta\tau\left[\left(k_\tau^l\right)_{\tau\le t}^{l\in L},\left(k_\tau^b\right)_{\tau\le t}^{b\in B}\right]=\sum_{\tau=0}^{t}\left[\sum_{l\in L}\sum_{i=k_\tau^l-\sum_{b\in B}q^l(b)+1}^{k_\tau^l}v_{i\tau}^l-\sum_{b\in B}c_{k_\tau^b\tau}^b\right]$.

To assess social welfare more accurately, we need to show that the difference between the valuations (costs) of shippers i and $i+1$ (carriers $j+1$ and j) is always bounded. By similar arguments in Huang et al. (2002), we can have the following lemma.

Lemma EC.4. *For all* $i=1,2,\ldots,m_t^l-1$, $\frac{1}{\varphi^l\left(m_\tau^l+1\right)}\le E\left[v_i^l-v_{i+1}^l\right]\le\frac{1}{\phi^l\left(m_\tau^l+1\right)}$; *and for all* $j=1,2,\ldots,n_\tau^b-1$, $\frac{1}{\delta^b\left(n_\tau^b+1\right)}\le E\left[c_{j+1}^b-c_j^b\right]\le\frac{1}{\lambda^b\left(n_\tau^b+1\right)}$; *where* ϕ^l, λ^b, φ^l, *and* δ^b *are defined as* $\phi^l=\min\left\{g^l(x):\underline{v}^l\le x\le\overline{v}^l\right\}$, $\lambda^b=\min\left\{g^b(x):\underline{c}^b\le x\le\overline{c}^b\right\}$, $\varphi^l=\max\left\{g^l(x):\underline{v}^l\le x\le\overline{v}^l\right\}$, *and* $\delta^b=\max\left\{g^b(x):\underline{c}^b\le x\le\overline{c}^b\right\}$.

Re-write $\widehat{V}_1\left[\left(K^l\right)^{l\in L},\left(K^b\right)^{b\in B}\right]$ as

$$\begin{aligned}\sum_{\tau=0}^{t}\widehat{V}_1\left[\left(k_\tau^l\right)_{\tau\le t}^{l\in L},\left(k_\tau^b\right)_{\tau\le t}^{b\in B}\right]=\sum_{\tau=0}^{t}\Bigg[&\sum_{l\in L}\left(v_{1\tau}^l-v_{2\tau}^l\right)+2\left(v_{2\tau}^l-v_{3\tau}^l\right)+\ldots+\left(k_\tau^l-1\right)\left(v_{\left(k_\tau^l-1\right)\tau}^l-v_{k_\tau^l\tau}^l\right)\\&+\sum_{b\in B}\left(c_{2\tau}^b-c_{1\tau}^b\right)+2\left(c_{3\tau}^b-c_{2\tau}^b\right)+\ldots+\left(k_\tau^b-1\right)\left(c_{\left(k_\tau^b\right)\tau}^b-c_{\left(k_\tau^b-1\right)\tau}^b\right)\\&+\left(\sum_{l\in L}k_\tau^l v_{k_\tau^l\tau}^l-\sum_{b\in B}k_\tau^b c_{k_\tau^b\tau}^b\right)\Bigg].\end{aligned}$$

By **Lemma EC.4**, we have

$$\begin{aligned}E\left[\sum_{\tau=0}^{t}V\frown_1\left[\left(k_\tau^l\right)_{\tau\le t}^{l\in L},\left(k_\tau^b\right)_{\tau\le t}^{b\in B}\right]\right]&\ge E\left[\sum_{\tau=0}^{t}\left[\sum_{l\in L}\frac{k_\tau^l\left(k_\tau^l-1\right)}{2\varphi^l(|m_\tau^l|+1)}+\sum_{b\in B}\frac{k_\tau^b\left(k_\tau^b-1\right)}{2\delta^b(|n_\tau^b|+1)}\right]\right]\\&=\underbrace{\sum_{l\in L}\frac{K^l\left(K^l-1\right)}{2\varphi^l(|\overline{m}^l|+1)}}_{A}+\underbrace{\sum_{b\in B}\frac{K^b\left(K^b-1\right)}{2\delta^b(|\overline{n}^b|+1)}}_{B}.\end{aligned}$$

As for the social welfare loss, we have

$$\begin{aligned}E\left[\sum_{\tau=0}^{t}\Delta\tau\left[\left(k_\tau^l\right)_{\tau\le t}^{l\in L},\left(k_\tau^b\right)_{\tau\le t}^{b\in B}\right]\right]&\le E\left[\sum_{\tau=0}^{t}\left[\sum_{l\in L}\left(v_{1\tau}^l-v_{k_\tau^l}^l\right)\sum_{b\in B}q^l(b)+\sum_{b\in B}\left(c_{k_\tau^b+1}^b-c_{k_\tau^b}^b\right)\right]\right]\\&\le E\left[\sum_{\tau=0}^{t}\left[\sum_{l\in L}\frac{\left(k_\tau^l-1\right)\sum_{b\in B}q^l(b)}{\phi^l(|m_\tau^l|+1)}+\sum_{b\in B}\frac{1}{\lambda^b(|n_\tau^b|+1)}\right]\right]\\&=\underbrace{\sum_{l\in L}\frac{\left(K^l-1\right)\sum_{b\in B}q^l(b)}{\phi^l(|\overline{m}^l|+1)}}_{C}+\underbrace{\sum_{b\in B}\frac{1}{\lambda^b(|\overline{n}^b|+1)}}_{D}.\end{aligned}$$

According to the definition of the efficiency loss, we have that

$$E[el(t)]\le E\left[\sum_{\tau=0}^{t}\frac{\Delta\tau}{\widehat{V}_1(I^t,J^t)}\right]=\frac{E\left[\sum_{\tau=0}^{t}\Delta\tau\left[\left(k_\tau^l\right)_{\tau\le t}^{l\in L},\left(k_\tau^b\right)_{\tau\le t}^{b\in B}\right]\right]}{E\left[\sum_{\tau=0}^{t}\widehat{V}_1\left[\left(k_\tau^l\right)_{\tau\le t}^{l\in L},\left(k_\tau^b\right)_{\tau\le t}^{b\in B}\right]\right]}=\frac{C+D}{A+B}.$$

Note that $\frac{C+D}{A+B}\le\max\left(\frac{C}{A},\frac{D}{B}\right)$. We can consider $\frac{C}{A}$ as the efficiency loss by removing shippers and $\frac{D}{B}$ as the efficiency loss by removing carriers. Then we have

$$E[el(t)] \le \max\left\{\max_{l\in L}\left[\frac{2\varphi^l\sum_{b\in B}q^l(b)}{\phi^l K^l}\right], \max_{b\in B}\left[\frac{2\delta^b}{K^b(K^b-1)\lambda^b}\right]\right\}$$

$$\le \max\left\{\frac{1}{\min_{l\in L}[K^l]}\max_{l\in L}\left[\frac{2\varphi^l\sum_{b\in B}q^l(b)}{\phi^l}\right], \frac{1}{\min_{b\in B}[K^b(K^b-1)]}\max_{b\in B}\left[\frac{2\delta^b}{\lambda^b}\right]\right\}$$

$$= O\left(\max\left\{\frac{1}{\min_{l\in L}[K^l]}, \frac{1}{\min_{b\in B}[K^b(K^b-1)]}\right\}\right).$$

We prove **Theorem 3**. ***Q.E.D.***

Proof of Theorem 4

Strong feasibility. Let $\left\{\left(\alpha_i^{t'}\right)_{i\in\vec{I}^t}, \left(\beta_j^{t'}\right)_{j\in\vec{J}^t}\right\}$ be the optimal solution to $\boldsymbol{IP3}(\vec{I}^t,\vec{J}^t,\boldsymbol{S})$, we have $\sum_{i\in\{i|l_i=l,i\in\vec{I}^t\}}\alpha_i^{t'} = \sum_{j\in\vec{J}^t}\beta_j^{t'}q^l(b_j) + S_l^t$ for each $l\in L$ from constraint (16). Let $B^{t'} = \left\{b\middle|\sum_{j\in\{j|b_j=b,j\in\vec{J}^t\}}\beta_j^{t'}\ge 1, b\in B\right\}$. For each bundle $b\in B^{t'}$, there are $\sum_{j\in\{j|b_j=b,j\in\vec{J}^t\}}\beta_j^{t'}-1$ asks. For each lane l, there are at least $\sum_{i\in\{i|l_i=l,i\in\vec{I}^t\}}\alpha_i^{t'} - \sum_{b\in B^{t'}}q^l(b) - (S_l-1)$ units of demand because $S_l = \max\{q_i|l_i=l, i\in I\}$. Then, for each lane $l\in L$, we have $\sum_{i\in\{i|l_i=l,i\in\vec{I}^t\}}\alpha_i^{t'} - \sum_{B^{t'}}q^l(b) - (S_l-1) = \sum_{j\in\vec{J}^t}\beta_j^{t'}q^l(b_j) - \sum_{b\in B^{t'}}q^l(b) + 1 = \sum_{b\in B^{t'}}\left(\sum_{j\in\{j|b_j=b,j\in\vec{J}^t\}}\beta_j^{t'}-1\right)q^l(b_j) + 1$.

So the demand exceeds supply and $\sum_{j\in\{j|b_j=b,j\in\vec{J}^t\}}y_j^t = \sum_{j\in\{j|b_j=b,j\in\vec{J}^t\}}\beta_j^{t'}-1$. From the allocation rule, we have $\sum_{n=1}^{|I_l^t|}x_n^t = \min\left\{\sum_{i\in I_l^t}q_i, \sum_{j\in J^t}y_j^t q^l(b_j)\right\}$ for each lane l in period t from Eq.(19). Then we must have $\sum_{n=1}^{|I_l^t|}x_n^t = \sum_{j\in J^t}y_j^t q^l(b_j) \le \sum_{i\in\{i|l_i=l,i\in\vec{I}^t\}}\alpha_i^{t'} - \sum_{B^{t'}}q^l(b) - (S_l-1) \le \sum_{i\in I_l^t}q_i$.

In each period t, the demand and supply are exactly balanced for each lane l. Thus, the OBDA-padding is *strongly* feasible.

Ex-post weak budget balance. Similar to the proof of **Theorem 1,** we can easily check that the OBDA-padding mechanism is *weakly* budget-balanced.

Ex-post individual rationality. For trading shipper i, it must be active. So we have $\widehat{v}_i > p_i$. If it trades in period t, we have $\widehat{v}_i > f_I^{l_i}(\vec{I}^t,\vec{J}^t)$. Trading shipper i pays $\pi_i^t = x_i^t\max\left\{p_i, f_I^{l_i}(\vec{I}^t,\vec{J}^t)\right\}$ in period t. Therefore, $\sum_{t\in[a_i,d_i]}x_i^t\widehat{v}_i > \sum_{t\in[a_i,d_i]}\pi_i^t$ and the trading shipper obtains non-negative utility from truthful bidding. By similar arguments as the proof of **Theorem 1** about individual rationality, we can show that the carriers can obtain positive utilities from truthful bidding. Thus, the OBDA-padding mechanism is individually rational. ***Q.E.D.***

Proof of Theorem 5.

We first show **Lemma EC.5** that the OBDA-padding mechanism is incentive-compatible in a single period.

Lemma EC.5. *The OBDA-padding mechanism is incentive-compatible in a single period.*

Proof. We first discuss the shipper side. For lane l, shipper i's buying price $f_I^{l_i}(I,J)$ is $\widehat{v}_{i_l^*}$, which is determined by shipper i_l^*'s bid price. For shipper i with $v_i < \widehat{v}_{i_l^*}$, $\widehat{v}_{i_l^*}$ keeps unchanged if it decreases its bid price, and it would get negative utility if it bids $\widehat{v}_i > \widehat{v}_{i_l^*}$. Then it has no incentive to lie. For shipper i with $v_i > \widehat{v}_{i_l^*}$, $\widehat{v}_{i_l^*}$ keeps unchanged if it increases its bid price, and it would not trade if it bids $\widehat{v}_i < \widehat{v}_{i_l^*}$. It also has no incentive to lie. For shipper i_l^* whose bid price determines $f_I^{l_i}(I,J)$, if it increases or decreases the bid price such that $f_I^{l_i}(I,J)$ is also equal to $\widehat{v}_{i_l^*}$, it would not trade. If it increases or decreases the bid price such that $f_I^{l_i}(I,J)$ is not equal to $\widehat{v}_{i_l^*}$, it would get a larger buying price than its valuation (overbidding) or not trade (underbidding). So it has no incentive to lie.

Due to supply padding, a shipper's demand may be partially met. Under the OBDA-padding mechanism, allocation is determined according to the bidding time, rather than the bid price. So shippers cannot get more allocation by misrepresenting the bid price.

In summary, bidding the valuation truthfully is a weakly dominant strategy for the shippers under the OBDA-padding mechanism in a single period. By similar arguments, we can also show that the carriers will ask their costs truthfully. ***Q.E.D.***

We next show **Lemma EC.6** that the agents will bid (ask) their true valuations (costs) under the OBDA-padding mechanism.

Lemma EC.6. *The agents will bid their valuations and costs truthfully under the OBDA-padding mechanism*.

Proof. If shipper i gets x_i^t units of the transport service in period t, its payment is $\pi_i^t = x_i^t\max\left\{p_i, f_I^{l_i}(\vec{I}^t,\vec{J}^t)\right\}$. If $d_i - a_i = K$, $p_i = 0$, which is independent of $\widehat{v}_i$. If $d_i - a_i < K$, then $p_i = \max_{\tau\in[\max(0,\widehat{d}_i-K),\widehat{a}_i-1]}f_I^{l_i}(\vec{I}^\tau\cup\{i\},\vec{J}^\tau)$. In each period $t\in[\max(0,\widehat{d}_i-K),\widehat{a}_i-1]$, $f_I^{l_i}(\vec{I}^t\cup\{i\},\vec{J}^t)$ is only dependent on the shipper set $\vec{I}^t\cup\{i\}$ and carrier set $\vec{J}^t$ rather than the agent sets $\vec{I}^\tau\cup\vec{J}^\tau$ in the previous periods ($\tau<t$) or subsequent periods ($\tau>t$). From **Lemma EC.5**, shipper i's weakly dominant strategy is to bid $\widehat{v}_i = v_i$ in a single period. So its weakly dominant strategy is to bid such $\widehat{v}_i = v_i$ in each period $\tau\in[\max(0,\widehat{d}_i-K),\widehat{a}_i-1]$. Thus, the shippers will bid their valuations truthfully under the OBDA-padding mechanism.

By similar arguments, we can also show that the carriers will ask their costs truthfully. ***Q.E.D.***

Lemma EC.7 shows that the agents will join the auction as soon as they arrive at the market, i.e., $\widehat{a}_i = a_i$ for each i and $\widehat{a}_j = a_j$ for each j.

Lemma EC.7. *The agents will not delay joining the auction under the OBDA-padding mechanism*.

Proof. We first discuss the shipper side. If $d_i - a_i = K$, shipper i will not delay joining the auction because p_i is zero, which is the lowest price it will get.

Then we consider the case where $d_i - a_i < K$. Shipper i can trade $\sum_{t=a_i}^{d_i} x_i^t$ units while bidding truthfully and its payment $\pi_i = \sum_{t=a_i}^{d_i} x_i^t \max\left\{\max_{\tau\in[\max(0,d_i-K),a_i-1]} f_I^{l_i}(\overline{I}^\tau \cup \{i\}, \overline{J}^\tau), \max_{\tau\in[a_i,d_i]} f_I^{l_i}(\overline{I}^\tau, \overline{J}^\tau)\right\}$. Because the earlier the bidding time is, the higher the priority of the transaction to be allocated. Shipper i cannot get more than $\sum_{t=a_i}^{d_i} x_i^t$ units by arriving late ($a_i < \widehat{a}_i$).

From **Lemma EC.6**, a shipper does not misrepresent its bid price to affect the pre-buying price in period t. From the pre-determined pricing rule, $f_I^{l_i}(I^t, J^t)$ for shipper i is determined by solving ***IP3***$(I^t, J^t, \boldsymbol{S})$:

$$\boldsymbol{IP3}\left(I^t, J^t, \boldsymbol{S}\right): \max_{(\alpha_i,\beta_j)} \widehat{V}_2(I^t, J^t) = \sum_{i\in I^t} \alpha_i \widehat{v}_i - \sum_{j\in J^t} \beta_j \widehat{c}_j \tag{54}$$

$$s.t. \sum_{i\in\{i|l_i=l,i\in J^t\}} \alpha_i = \sum_{j\in J^t} \beta_j q^l(b_j) + S_l, \forall l \in L \tag{55}$$

$$\alpha_i \le q_i - \sum_{\tau=\widehat{a}_i}^{t-1} x_i^\tau, \forall i \in I^t \tag{56}$$

$$\alpha_i \in Z_0^+, \forall i \in I^t \tag{57}$$

$$\beta_j \in \{0,1\}, \forall j \in J^t \tag{58}$$

If shipper i' bids $a_{i'} < \widehat{a}_{i'}$, we have that $q_{i'} - \sum_{\tau=\widehat{a}_{i'}}^{t-1} x_{i'}^\tau$ is not less than $q_{i'} - \sum_{\tau=a_{i'}}^{t-1} x_{i'}^\tau$ in constraint (56). So we will get more transactions ($\sum_{i\in\{i|l_i=l_{i'},i\in I^t\}} \alpha_i'$) being allocated in the optimal solution to ***IP3***$(I^t, J^t, \boldsymbol{S})$. From the pre-determined pricing rule, shipper i' will get a buying price $f_I^{l_{i'}}(\overline{I}^t, \overline{J}^t)$ no less than that as bidding truthfully. So the shipper prefers to get its allocation as early as possible. According to the allocation rule that the earlier the bidding time is, the higher the priority of the transaction to be allocated, $a_{i'}$ is the earliest time shipper i' can arrive. So shipper i' will not delay participation in the auction.

In summary, the shippers will not delay joining the auction under the OBDA-padding mechanism. By similar arguments, we can also show that the carriers will not delay joining the auction either. ***Q.E.D.***

Lemma EC.8 shows that the agents will not leave the market early, i.e., $\widehat{d}_i = d_i$ for each i and $\widehat{d}_j = d_j$ for each j.

Lemma EC.8. *The agents will not leave the market early under the OBDA-padding mechanism.*

Proof. We first discuss the shipper side. If $d_i - a_i = K$, shipper i will not leave the market early because p_i is zero, which is the lowest price it will pay.

If shipper i bids $\widehat{d}_i < d_i$, it will lose the trade opportunity from period $\widehat{d}_i$ to period d_i. Even if shipper i with $\widehat{d}_i < d_i$ gets the same total allocation $\sum_{t=a_i}^{d_i} x_i^t$ as bidding truthfully, its payment will not be lower than that as biding $\widehat{d}_i = d_i$ because $\max_{\tau\in[\max(0,\hat{d}_i-K),\hat{d}_i-1]} f_I^{l_i}(\overline{I}^\tau \cup \{i\}, \overline{J}^\tau) \ge \max_{\tau\in[\max(0,d_i-K),\hat{a}_i-1]} f_I^{l_i}(\overline{I}^\tau \cup \{i\}, \overline{J}^\tau)$. So shipper i cannot obtain a higher utility by early departure ($\widehat{d}_i < d_i$). By similar arguments, we can also show that a carrier cannot get higher utility by leaving early. Then we prove the lemma. ***Q.E.D.***

From **Lemmas EC.5–8**, the OBDA-padding mechanism is incentive-compatible. ***Q.E.D.***

Proof of Theorem 6

The lower bound on the *ex-post* maximum social welfare in the case with multi-unit demand can be written as,

$$\begin{aligned}
\sum_{\tau=0}^{t} \widehat{V}_1 \left[\left(k_\tau^l\right)_{\tau\le t}^{l\in L}, \left(k_\tau^b\right)_{\tau\le t}^{b\in B}\right] &= \sum_{\tau=0}^{t} \left[\sum_{l\in L}\sum_{i=1}^{k_\tau^l} \left(v_{i\tau}^l - v_{k_\tau^l\tau}^l\right) q_{i\tau}^l + \sum_{b\in B}\sum_{j=1}^{k_\tau^b} \left(c_{k_\tau^b\tau}^b - c_{j\tau}^b\right) + \left(\sum_{l\in L} \alpha_{i\tau}^{\tau'} v_{k_\tau^l\tau}^l - \sum_{b\in B} k_\tau^b c_{k_\tau^b\tau}^b\right)\right] \\
&= \sum_{\tau=0}^{t} \left[\sum_{l\in L} \left[\left(v_{1\tau}^l - v_{2\tau}^l\right) q_{1\tau}^l + \left(v_{2\tau}^l - v_{3\tau}^l\right)\left(q_{1\tau}^l + q_{2\tau}^l\right) + \ldots + \left(v_{\left(k_\tau^l-1\right)\tau}^l - v_{k_\tau^l\tau}^l\right) \sum_{i=1}^{k_\tau^l-1} q_{i\tau}\right]\right. \\
&\quad + \sum_{b\in B} \left[\left(c_{2\tau}^b - c_{1\tau}^l\right) + 2\left(c_{3\tau}^b - c_{2\tau}^b\right) + \ldots + \left(k_\tau^b - 1\right)\left(c_{k_\tau^b\tau}^b - c_{\left(k_\tau^b-1\right)\tau}^b\right)\right] \\
&\quad \left. + \left(\sum_{l\in L} \alpha_{i\tau}^{\tau'} v_{k_\tau^l\tau}^l - \sum_{b\in B} k_\tau^b c_{k_\tau^b\tau}^b\right)\right].
\end{aligned}$$

If $d - a = K$ for any agent, we have that $\overline{I}^\tau = I^\tau$ and $\overline{J}^\tau = J^\tau$. Then the welfare loss is caused by the padding and the removed transactions as follows,

$$\sum_{\tau=0}^{t}\Delta\tau\left[\left(k_\tau^l\right)_{\tau\le t}^{l\in L},\left(k_\tau^b\right)_{\tau\le t}^{b\in B}\right]\le\sum_{\tau=0}^{t}\left[\sum_{l\in L}\sum_{i=1}^{\widetilde{k}_\tau^l}q_{i\tau}v_{i\tau}^l-\sum_{b\in B}c_{k_\tau^b\tau}^b\right]$$

$$\le\sum\nolimits_{\tau=0}^{t}\left[\sum\nolimits_{l\in L}\left(v_{1\tau}^l-v_{k_\tau^l}^l\right)\left[S_l+\sum\nolimits_{b\in B}q^l(b)\right]+\sum\nolimits_{b\in B}\left(c_{\left(k_\tau^b+1\right)\tau}^b-c_{k_\tau^b\tau}^b\right)\right].$$

where $\widetilde{k}_\tau^l=\min\left\{i'\middle|\sum_{i=1}^{i'}q_{i\tau}\ge\sum_{b\in B'}q^l(b)+S_l,i'\in I_t^l\right\}$.

By **Lemma EC.4,** we have

$$E\left[\sum_{\tau=0}^{t}\widehat{V}_1\left[\left(k_\tau^l\right)_{\tau\le t}^{l\in L},\left(k_\tau^b\right)_{\tau\le t}^{b\in B}\right]\right]\ge E\left[\sum_{\tau=0}^{t}\left[\sum_{l\in L}\frac{k_\tau^l\left(k_\tau^l-1\right)\overline{q}^l}{2\varphi^l(|m_\tau^l|+1)}+\sum_{b\in B}\frac{k_\tau^b\left(k_\tau^b-1\right)}{2\delta^b(|n_\tau^b|+1)}\right]\right]$$

$$=\underbrace{\sum\nolimits_{l\in L}\frac{K^l(K^l-1)\overline{q}^l}{2\varphi^l(|\overline{m}^l|+1)}}_{A}+\underbrace{\sum\nolimits_{b\in B}\frac{K^b(K^b-1)}{2\delta^b(|\overline{n}^b|+1)}}_{B}.$$

In addition, we have that

$$E\left[\sum_{\tau=0}^{t}\Delta\tau\left[\left(k_\tau^l\right)_{\tau\le t}^{l\in L},\left(k_\tau^b\right)_{\tau\le t}^{b\in B}\right]\right]\le E\left[\sum_{\tau=0}^{t}\left[\sum_{l\in L}\left(v_{1\tau}^l-v_{k_\tau^l}^l\right)\left[S_l+\sum_{b\in B}q^l(b)\right]+\sum_{b\in B}\left(c_{\left(k_\tau^b+1\right)\tau}^b-c_{k_\tau^b\tau}^b\right)\right]\right]$$

$$\le E\left[\sum\nolimits_{\tau=0}^{t}\left[\sum\nolimits_{l\in L}\frac{\left(k_\tau^l-1\right)\left[S_l+\sum_{b\in B}q^l(b)\right]}{\phi^l(|m_\tau^l|+1)}+\sum\nolimits_{b\in B}\frac{1}{\lambda^b(|n_\tau^b|+1)}\right]\right]$$

$$=\underbrace{\sum\nolimits_{l\in L}\frac{\left(K^l-1\right)\left[S_l+\sum_{b\in B}q^l(b)\right]}{\phi^l(|\overline{m}^l|+1)}}_{C}+\underbrace{\sum\nolimits_{b\in B}\frac{1}{\lambda^b(|\overline{n}^b|+1)}}_{D}.$$

Note that $\frac{C+D}{A+B}\le\max\left(\frac{C}{A},\frac{D}{B}\right)$. We can consider $\frac{C}{A}$ as the efficiency loss by removing shippers and $\frac{D}{B}$ as the efficiency loss by removing carriers. Then we have

$$E[el(t)]\le\max\left\{\max_{l\in L}\left[\frac{2\varphi^l\left[S_l+\sum_{b\in B}q^l(b)\right]}{\phi^lK^l\overline{q}^l}\right],\max_{b\in B}\left[\frac{2\delta^b}{K^b\left(K^b-1\right)\lambda^b}\right]\right\}$$

$$\le\max\left\{\max_{l\in L}\left[\frac{S_l+\sum_{b\in B}q^l(b)}{K^l\overline{q}^l}\right]\max_{l\in L}\left[\frac{2\varphi^l}{\phi^l}\right],\frac{1}{\min_{b\in B}\left[K^b\left(K^b-1\right)\right]}\max_{b\in B}\left[\frac{2\delta^b}{\lambda^b}\right]\right\}$$

$$=O\left(\max\left\{\max_{l\in L}\left[\frac{S_l+\sum_{b\in B}q^l(b)}{K^l\overline{q}^l}\right],\frac{1}{\min_{b\in B}\left[K^b\left(K^b-1\right)\right]}\right\}\right).$$

We prove **Theorem 6**. ***Q.E.D.***

Proof of Theorem 7

We first prove that the OBDA-SMP mechanism is incentive-compatible. We need the following lemma.

Lemma EC.9. *The OBDA-SMP mechanism is incentive-compatible in a single period.*

Proof. We first discuss the shipper side. For shipper i wants demand bundle o, its buying price is $\widehat{v}_{i_o^*}(o)$. If $v_i(o)>\widehat{v}_{i_o^*}(o)$, overbidding ($\widehat{v}_i(o)>v_i(o)$) will induce the same buying price as it bids v_i; underbidding ($\widehat{v}_i(o)<v_i(o)$) may induce it to lose the trade ($\widehat{v}_i(o)\le\widehat{v}_{i_o^*}(o)<v_i(o)$). And even if it wins the trade ($\widehat{v}_{i_o^*}(o)<\widehat{v}_i(o)<v_i(o)$), the buying price is the same as it bids truthfully. If $v_i(o)\le\widehat{v}_{i_o^*}(o)$, shipper i does not trade. Underbidding ($\widehat{v}_i(o)<v_i(o)$) will not make it trade either. Overbidding ($\widehat{v}_i(o)>v_i(o)$) will not make it trade ($\widehat{v}_{i_o^*}(o)>\widehat{v}_i(o)>v_i(o)$) or will cause negative utility ($\widehat{v}_i(o)>(o)\widehat{v}_{i_o^*}\ge v_i(o)$).Therefore, the shippers will submit the true valuations for their demand bundles.

By similar arguments as above, we can show that the carriers will also bid their costs truthfully. Then we prove **Lemma EC.9**. ***Q.E.***

D.

From **Lemma EC.9**, by similar arguments as the proof **Theorem 2**, we can prove the incentive compatibility of the OBDA-SMP mechanism. By similar arguments as the proof of **Theorem 1**, we can easily check that the OBDA-SMP mechanism is *strongly* feasible, *ex-post weakly* budget-balanced, and *ex-post* individually rational. ***Q.E.D.***

Proof of Proposition 1

Let $\eta_i(e) = \gamma(e) \times \alpha_i(e)$ and $\kappa_j(e) = \gamma(e) \times \beta_j(e)$. The relationship among $\eta_i(e)$, $\gamma(e)$ and $\alpha_i(e)$ is shown in Table A.1.

Table A1
The relationship among $\eta_i(e)$, $\gamma(e)$ and $\alpha_i(e)$.

$\eta_i(e)$	$\gamma(e)$	$\alpha_i(e)$
0	0	0
0	1	0
0	0	1
1	1	1

From Table A.1, we have the following results: (i) $\eta_i(e) \le \gamma(e)$; (ii) $\eta_i(e) \le \alpha_i(e)$; (iii) $\gamma(e) + \alpha_i(e) - \eta_i(e) \le 1$; and (iv) $\eta_i(e) \in \{0,1\}$. Because $\gamma(e)$ denotes whether the matching pattern e is selected and $\alpha_i(e)$ denotes whether shipper i trades under the matching pattern e, $\alpha_i(e) = 0$ is an optimal solution if $\gamma(e) = 0$. Then we have that $\alpha_i(e) \le \gamma(e)$. By the transitivity of the inequality, we can replace $\eta_i(e) \le \gamma(e)$ with $\alpha_i(e) \le \gamma(e)$. Thus $\gamma(e) \times \alpha_i(e)$ for each $i \in I$ and each $e \in E$ can be transformed into the linear constraints (43)-(45) and (49).

By the similar linearization technique, we can transform $\gamma(e) \times \beta_j(e)$ for each $j \in J$ and each $e \in E$ into constraints (46)-(48) and (50).

Then the objective function of ***INLP1***(I, J), $\sum_{e\in E}\gamma(e)\left[\sum_{i\in I}\alpha_i(e)v_i(o_i) - \sum_{j\in J}\beta_j(e)c_j\right]$, can be rewritten as $\sum_{e\in E}\left[\sum_{i\in I}\eta_i(e)v_i(o_i) - \sum_{j\in J}\kappa_j(e)c_j\right]$.

In summary, we show that ***INLP1***(I,J) can be reduced to ***IP6***(I,J) and the optimal solution to ***IP6***(I,J) is also an optimal solution to ***INLP1***(I,J). ***Q.E.D.***

Complementary numerical studies on a U.S. Dataset

In this subsection, we consider a transport network consisting of five hub cities in the U.S. (i.e., Chicago, Houston, Los Angeles, New York, and Philadelphia). In each simulation test, the agents trade transport services on ten lanes that are randomly generated among these five cities. Other parameters are set as those in Section 6. We repeat the randomized simulation test 20 times for each set of parameter settings and report the average values.

Table A.2 reports the performance of the OBDA, OBDA-padding, and OBDA-SMP mechanisms at different market sizes.

Table A2
The performance of the mechanisms at different market sizes.

	OBDA mechanism			OBDA-padding mechanism			OBDA-SMP mechanism		
	Efficiency	Agents	Platform	Efficiency	Agents	Platform	Efficiency	Agents	Platform
10 bundle types									
20	88.68 %	67.57 %	$271,903	83.49 %	66.09 %	$256,611	69.70 %	60.72 %	$201,288
30	93.32 %	74.21 %	$333,888	90.47 %	73.49 %	$312,049	79.96 %	67.38 %	$277,187
40	95.43 %	78.81 %	$388,762	92.63 %	77.24 %	$404,047	80.40 %	69.57 %	$379,527
13 bundle types									
20	91.47 %	71.59 %	$351,723	87.75 %	70.65 %	$309,459	77.91 %	65.15 %	$271,021
30	94.97 %	77.81 %	$411,603	92.91 %	77.54 %	$393,383	83.73 %	71.62 %	$368,510
40	96.61 %	81.60 %	$434,707	94.88 %	81.09 %	$446,198	85.91 %	76.16 %	$413,584
16 bundle types									
20	92.93 %	73.83 %	$392,734	89.91 %	72.43 %	$386,579	81.98 %	67.56 %	$333,055
30	95.90 %	79.92 %	$464,279	93.88 %	79.00 %	$429,876	86.05 %	73.74 %	$391,721
40	96.85 %	82.36 %	$597,730	95.87 %	83.12 %	$494,046	89.69 %	77.96 %	$488,428

From Table A.2, we have the following findings: (i) among the three mechanisms, the OBDA mechanism is the most efficient, followed by the OBDA-padding mechanism, and finally the OBDA-SMP mechanism; (ii) the efficiencies of the three mechanisms increase with the market size; (iii) the mechanisms tend to allocate more social welfare to the agents; and (iv) as the market size increases, the agents' share of social welfare and the platform's profit increase. These findings are consistent with that in Subsection 6.2.

Table A.3 reports the performance of the OBDA, OBDA-padding, and OBDA-SMP mechanisms with different agents' patience.

Table A3
The performance of the mechanisms with different agents' patience.

K	OBDA mechanism			OBDA-padding mechanism			OBDA-SMP mechanism		
	Efficiency	Agents	Platform	Efficiency	Agents	Platform	Efficiency	Agents	Platform
0	99.25 %	88.39 %	$211,399	96.65 %	87.96 %	$207,132	90.89 %	84.41 %	$204,483
1	94.97 %	77.81 %	$411,603	92.91 %	77.54 %	$393,383	83.73 %	71.62 %	$368,510
2	87.88 %	63.78 %	$627,504	85.08 %	62.31 %	$605,222	70.49 %	56.31 %	$476,304
3	81.29 %	56.96 %	$661,999	77.15 %	55.86 %	$652,936	69.30 %	51.42 %	$453,440

From Table A.3, we have the following findings: (i) the efficiency of the mechanisms decreases with the agents' patience bound; (ii) the efficiency of the mechanisms is very efficient when the agents are impatient; (iii) as the agents' patience increases, the incremental profit of the platform tends to first increase and then decrease. These findings are consistent with that in Subsection 6.3.

In summary, although we consider a dataset generated on the U.S. transport network, we can get similar findings to that in Section 6. Therefore, our findings from the numerical studies are general but not limited to the Chinese scenario.

Robustness check on the number of experiments

In this subsection, we conduct a robustness check on the number of experiments. Specifically, we consider 10, 20, and 30 randomized simulation tests for each set of parameter settings, respectively. We use the average result of 20 experiments as a benchmark. Table A.4 shows the percentage differences between the average results of 10 experiments and the benchmark results at different market sizes, where $percentagedifference = \frac{(theaverageresultof10experiment - benchmarkresult)}{benchmarkresult} \times 100\%$.

Table A4
The percentage differences between the average results of 10 experiments and the benchmarks at different market sizes.

	OBDA mechanism			OBDA-padding mechanism			OBDA-SMP mechanism		
	Efficiency	Agents	Platform	Efficiency	Agents	Platform	Efficiency	Agents	Platform
10 bundle types									
20	−0.30 %	−0.17 %	−2.42 %	0.58 %	0.86 %	0.81 %	0.15 %	0.56 %	−0.23 %
30	−0.36 %	−0.89 %	2.32 %	0.41 %	0.18 %	−0.05 %	0.81 %	0.81 %	−0.25 %
40	−0.13 %	−0.98 %	−1.20 %	0.08 %	−0.28 %	−0.49 %	−0.08 %	−0.40 %	0.74 %
13 bundle types									
20	0.40 %	−0.57 %	0.45 %	0.09 %	−0.25 %	0.87 %	−0.17 %	0.72 %	−2.89 %
30	0.18 %	0.81 %	1.67 %	−0.21 %	−0.10 %	−0.32 %	0.77 %	−0.12 %	1.37 %
40	−0.07 %	−0.19 %	−1.86 %	0.32 %	0.93 %	−2.77 %	0.11 %	−0.06 %	−1.43 %
16 bundle types									
20	−0.31 %	−0.61 %	1.47 %	0.41 %	0.38 %	−3.51 %	0.18 %	0.45 %	0.25 %
30	−0.02 %	−0.08 %	0.16 %	−0.33 %	0.19 %	0.71 %	−1.00 %	−0.17 %	2.95 %
40	−0.01 %	0.20 %	−0.18 %	−0.17 %	0.35 %	−0.83 %	−0.66 %	1.25 %	1.25 %

Table A.5 shows the percentage differences between the average results of 30 experiments and the benchmark results at different market sizes.

Table A5
The percentage differences between the average results of 30 experiments and the benchmark at different market sizes.

	OBDA mechanism			OBDA-padding mechanism			OBDA-SMP mechanism		
	Efficiency	Agents	Platform	Efficiency	Agents	Platform	Efficiency	Agents	Platform
10 bundle types									
20	88.95 %	66.91 %	$92,472	−0.24 %	0.63 %	0.16 %	−0.39 %	0.21 %	0.67 %
30	−0.19 %	0.32 %	0.27 %	−0.43 %	−0.82 %	0.34 %	0.23 %	−0.78 %	0.93 %
40	0.07 %	0.73 %	−1.11 %	0.03 %	0.54 %	0.01 %	−0.05 %	−0.19 %	1.13 %
13 bundle types									
20	−0.17 %	−0.28 %	−0.83 %	−0.03 %	−0.59 %	1.45 %	0.77 %	0.03 %	0.98 %
30	0.19 %	0.49 %	1.00 %	−0.34 %	−0.38 %	0.56 %	−0.20 %	0.23 %	−1.11 %
40	−0.03 %	−0.25 %	0.65 %	0.00 %	−0.20 %	1.25 %	0.05 %	0.41 %	1.26 %
16 bundle types									
20	0.05 %	0.27 %	−1.35 %	0.10 %	0.13 %	0.67 %	−0.62 %	0.45 %	1.37 %
30	0.05 %	0.04 %	−1.12 %	0.12 %	0.14 %	−1.51 %	−0.01 %	−0.65 %	−1.87 %
40	0.02 %	0.15 %	−1.23 %	0.01 %	0.05 %	−1.43 %	−0.71 %	−0.41 %	−0.95 %

Tables A.6 and A.7 show the percentage differences between the average results of 10 and 30 experiments and the benchmark results with different agents' patience, respectively.

Table A6
The percentage differences between the average results of 10 experiments and the benchmark with different agents' patience.

K	OBDA mechanism			OBDA-padding mechanism			OBDA-SMP mechanism		
	Efficiency	Agents	Platform	Efficiency	Agents	Platform	Efficiency	Agents	Platform
0	0.03 %	0.08 %	0.33 %	0.09 %	0.59 %	−1.51 %	−0.64 %	0.38 %	0.64 %
1	0.18 %	0.81 %	1.67 %	−0.21 %	−0.10 %	−0.32 %	0.77 %	−0.12 %	1.37 %
2	−0.63 %	−1.65 %	1.95 %	0.36 %	−0.54 %	2.74 %	0.06 %	−0.93 %	2.57 %
3	0.00 %	−0.59 %	−1.28 %	0.91 %	1.02 %	−2.66 %	0.43 %	2.03 %	0.48 %

Table A7
The percentage differences between the average results of 30 experiments and the benchmark with different agents' patience.

K	OBDA mechanism			OBDA-padding mechanism			OBDA-SMP mechanism		
	Efficiency	Agents	Platform	Efficiency	Agents	Platform	Efficiency	Agents	Platform
0	−0.04 %	−0.42 %	1.72 %	−0.08 %	−0.32 %	−1.04 %	−0.15 %	0.09 %	−1.10 %
1	0.19 %	0.49 %	1.00 %	−0.34 %	−0.38 %	0.56 %	−0.20 %	0.23 %	−1.11 %
2	0.20 %	0.61 %	0.27 %	−0.09 %	−0.24 %	−1.67 %	0.18 %	0.64 %	−1.83 %
3	0.26 %	0.10 %	0.43 %	−0.16 %	−0.25 %	0.17 %	−0.20 %	−0.38 %	0.98 %

From Tables A.4-A.7, when the number of experiments is greater than 10, it has a very small effect on the realized average mechanism efficiency and agents' share of social welfare. The effect of the experiment number on the realized average platform profit is slightly larger, but still less than 3 %. In summary, the number of experiments has a very limited effect on our average experimental results in Section 6, and thus our findings are robust.

Data availability

Data will be made available on request.

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